

# Revisiting the Concrete Security of FALCON-type Signatures

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**Abstract.** FALCON is a selected signature scheme in the NIST post-quantum standardization. It is an efficient instantiation of the GPV framework over NTRU lattices. While the GPV framework comes with an elegant security proof in theory, FALCON had no formal proof involving concrete parameters for a long time. Until recently, Fouque et al. initiate the concrete security analysis of FALCON-type signatures. They give a formal proof of FALCON<sup>+</sup>, a minor modification of FALCON, in the random oracle model, whereas they claim that FALCON<sup>+</sup>-512 barely achieves the claimed 120-bit security for plain unforgeability. Furthermore, they show that standard reductions for strong unforgeability are vacuous for FALCON parameters, necessitating the introduction of a new, non-standard assumption.

In this work, we revisit the concrete security analysis of FALCON-type signatures and present positive results. We develop improved analytic tools by leveraging the profile of the NTRU trapdoor bases. This eliminates the security loss for both FALCON<sup>+</sup>-512 and FALCON<sup>+</sup>-1024 in the case of plain unforgeability. We also apply our new analysis to the recent weak-smoothness variant FALCON<sup>ws</sup> (Zhang et al. Asiacrypt 2025) that admits smaller parameters than FALCON under a non-standard assumption. As a result, we propose new parameters for FALCON<sup>ws</sup> allowing for provable security under standard assumptions and signature size 17.8% (resp. 12.8%) smaller than that of FALCON-512 (resp. FALCON-1024) simultaneously. Moreover, we give a refined strong unforgeability security proof by replacing the worst-case analysis with a probabilistic analysis, which leads to a substantial increase in concrete security. Based on this, we show that by using a tighter Gaussian sampler, e.g. the one in FALCON<sup>ws</sup>, FALCON-type signatures can achieve concrete security for strong unforgeability closely consistent with the claimed security level while keeping the compact size.

**Keywords:** FALCON · Rényi Divergence · Security Proof · Lattice-Based Cryptography ·

# 1 Introduction

FALCON [PFH<sup>+</sup>22] is one of three digital signature algorithms selected by NIST for post-quantum cryptography (PQC) standardization. It offers excellent efficiency, particularly a significant advantage in terms of size, among them. FALCON is based on the celebrated GPV framework for lattice-based hash-and-sign signatures [GPV08] and adopts a set of efficiency-oriented designs, e.g. the nearly-optimal NTRU trapdoor [DLP14], a ring-efficient Gaussian sampler [DP16] and well-analyzed parameters [Pre17]. The security assumptions of FALCON are standard computational hard problems over NTRU lattices, which have been extensively studied. With the post-quantum standardization and migration underway, it is crucial to investigate the security of FALCON not only in a pure algorithmic cryptanalysis sense, but also with respect to provable security.

The GPV framework is quite elegant from a theoretical standpoint. It has a tight and arguably easy security proof in both the random oracle model (ROM) [GPV08] and the quantum random oracle model (QROM) [BDF<sup>+</sup>11], in contrast with Fiat-Shamir type constructions, e.g. another NIST winning signature scheme Dilithium [LDK<sup>+</sup>22]. However, such security analysis becomes complicated when it comes to concrete parameters: some arguments in the original proof could even be invalidated. The concrete security analysis of FALCON remains largely unexplored for quite a long time, despite FALCON being selected for standardization.

Until recently, Gajland, Janneck and Kiltz present the first concrete security analysis of FALCON-type signatures [FGdG<sup>+</sup>25]. To do so, they introduce a minor but crucial modification of FALCON, termed FALCON<sup>+</sup>, in which the signing algorithm picks a fresh salt within the repeat loop until a preimage with a norm below the bound  $\beta$  is found. This modification allows for a formal proof in the ROM. By leveraging elaborate Rényi arguments and recent proof techniques [BRTZ24], some concrete security bounds of FALCON<sup>+</sup> are calculated. Unfortunately, their results still do not provide compelling evidence to support the concrete security of FALCON-type signatures. In the case of plain unforgeability, while they confirmed the claimed security level for the FALCON-1024 parameter set, they showed a loss of 7 bits in provable security for FALCON<sup>+</sup>-512. Crucially, in the case of strong unforgeability, their analysis reveals that the traditional reduction from the Short Integer Solution (SIS) problem with norm  $2\beta$  is trivial for FALCON parameters. To circumvent this, they reduced strong unforgeability to a second-preimage version of the inhomogeneous SIS problem ( $t$ - $\mathcal{R}$ -SPISIS)<sup>6</sup>. Consequently, the security guarantee for strong unforgeability relies on the non-standard assumption that  $t$ - $\mathcal{R}$ -SPISIS is as hard as standard SIS.

The standardization also promoted the exploration of the design space to improve various aspects of PQC schemes. A number of FALCON variants and

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<sup>6</sup> The  $t$ - $\mathcal{R}$ -SPISIS problem was introduced in a major revision of [FGdG<sup>+</sup>25] published on November 6, 2025. This updated version includes two additional authors (Fouque and de Groote). The old version 20250226:153603 is cited as [GJK24] in this work.

modifications have emerged [CPS+20,EFG+22,ETWY22,ENS+23,LTYZ25] in the past few years. Very recently, Zhang et al. propose a variant of FALCON in [ZJR+25] achieving smaller size and higher security. Their FALCON variant works with the so-called weak smoothness condition in which the closeness parameter associated with  $\eta_\epsilon(\mathbb{Z}^n)$  is  $\epsilon = O(1)$ , hence we refer to it as  $\text{FALCON}^{\text{ws}}$ . The overly large  $\epsilon$  of  $\text{FALCON}^{\text{ws}}$  forbids the original GPV security proof: the adversary cannot program the RO, as the simulated syndrome is no longer statistically indistinguishable from uniform. To address this issue, Zhang et al. resorted to one additional non-standard hardness assumption to ensure the simulated syndrome is computationally indistinguishable from uniform. While they conducted some cryptanalysis on the new assumption to claim no impact on the concrete security, removing such a non-standard assumption while maintaining the gain in size could be an interesting problem.

### 1.1 Our Contributions.

In this work, we revisit the security analysis of FALCON-type signatures. We obtain some positive results. Our contributions are mainly threefold.

**Sharper Security Bounds Using the NTRU Basis Profile.** We present a refined concrete security analysis for FALCON that significantly tightens the provable bit security. Unlike the prior worst-case analysis in [FGdG+25], which conservatively estimated the Rényi loss for all possible bases, our approach incorporates the geometric profile of each specific basis generated by FALCON’s key generation. This basis-aware evaluation yields instance-specific security bounds that are substantially tighter in practice. In our extensive evaluation over 1,000 distinct basis instances, the Rényi loss are bounded by 0.2 bits in all cases. An overview of the provable bit security is shown in Table 1. Notice that the larger security margin of  $\text{FALCON}^+-1024$  compensates for the 8 bits Rényi losses. In fact, our analysis shows that the resulting Rényi loss for  $\text{FALCON}^+-1024$  is below 0.2 bits as well.

**Table 1.** Provable bit security levels of plain unforgeability for FALCON.

| Scheme                 | [FGdG+25] | Ours |
|------------------------|-----------|------|
| $\text{FALCON}^+-512$  | 113       | 120  |
| $\text{FALCON}^+-1024$ | 256       | 256  |

**On the Provable Security of  $\text{FALCON}^{\text{ws}}$ .** To improve the bandwidth efficiency of FALCON, Zhang et al. [ZJR+25] proposed a compact variant  $\text{FALCON}^{\text{ws}}$ , where the closeness parameter  $\epsilon = O(1) \gg 1/\sqrt{Q_s \lambda}$  for the maximum number of signing queries  $Q_s$  and security parameter  $\lambda$ . While this approach significantly improves bandwidth and security, their security proof relies on a non-standard

assumption. Furthermore, the Rényi divergence-based analysis from [FGdG<sup>+</sup>25] becomes ineffective here, as the large  $\epsilon$  causes the Rényi loss from the RO simulation to grow prohibitively. Removing this non-standard assumption remains an open challenge. In this work, we address this challenge by a refined smoothing parameter bound. We establish the maximum permissible value of  $\epsilon$  that preserves provable security under standard assumptions, albeit in a setting that differs from that of [ZJR<sup>+</sup>25]. The resulting parameter sets achieve a balanced trade-off between bandwidth and security. A comparative summary of our new parameter sets against prior designs is provided in Table 2.

**Table 2.** Comparison of FALCON variants in sizes (byte) and bit security.

|                            | PK Size | Sig. Size | Standard Assumption |
|----------------------------|---------|-----------|---------------------|
| FALCON <sup>+</sup> -512   | 897     | 666       | ✓                   |
| FALCON <sup>ws</sup> -512  | 640     | 474       | ×                   |
| Ours                       | 704     | 547       | ✓                   |
| FALCON <sup>+</sup> -1024  | 1793    | 1280      | ✓                   |
| FALCON <sup>ws</sup> -1024 | 1408    | 979       | ×                   |
| Ours                       | 1536    | 1115      | ✓                   |

**Refined Analysis of Strong Unforgeability.** As shown in [GJK24], the original  $2\beta$  norm bound incurred a 25-bit loss for FALCON<sup>+</sup>-512, while rendering the security guarantee for FALCON<sup>+</sup>-1024 void since  $2\beta > q$ . The updated version in [FGdG<sup>+</sup>25] circumvents this problem by adopting a non-standard assumption. We introduce a refined norm bound of  $\sqrt{2}\beta$ , motivated by the high likelihood of near-orthogonality between the forgery and simulated signature in the ROM. This modification reduces the security loss and enables strong unforgeability for both security levels. We further tighten the reduction by co-optimizing  $\epsilon$  and the auxiliary parameter  $L$ , a technique originating from [BRTZ24] and employed in [FGdG<sup>+</sup>25], for which we present an optimized strategy for its selection. As summarized in Table 3, our approach significantly improves the concrete strong unforgeability of FALCON compared to [GJK24, FGdG<sup>+</sup>25]. In addition, we propose new parameter sets for FALCON<sup>ws</sup> that achieve SUF-CMA security under standard assumptions. See Table 4 for more details.

**Table 3.** Comparison of provable bit security levels of strong unforgeability for  $\text{FALCON}^+$ .

|                        | $\text{FALCON}^+$ -512 | $\text{FALCON}^+$ -1024 | Standard Assumption |
|------------------------|------------------------|-------------------------|---------------------|
| [GJK24]                | 89                     | —                       | ✓                   |
| [FGdG <sup>+</sup> 25] | 113                    | 256                     | ×                   |
| Ours                   | 107                    | 251                     | ✓                   |

**Table 4.** Comparison of provable bit security levels of strong unforgeability for our revised  $\text{FALCON}^{\text{ws}}$  and  $\text{FALCON}^+$ .

|                                            | PK Size | Sig. Size | Security | Standard Assumption |
|--------------------------------------------|---------|-----------|----------|---------------------|
| $\text{FALCON}^+$ -512                     | 897     | 666       | 113      | ×                   |
| $\text{FALCON}^{\text{ws}}$ -512 with SUF  | 768     | 576       | 114      | ✓                   |
| $\text{FALCON}^+$ -1024                    | 1793    | 1280      | 256      | ×                   |
| $\text{FALCON}^{\text{ws}}$ -1024 with SUF | 1664    | 1179      | 265      | ✓                   |

## 2 Preliminaries

Let  $\mathbb{R}$ ,  $\mathbb{Z}$  and  $\mathbb{N}$  denote the set of real numbers, integers and natural numbers respectively. Vectors (resp. matrices) are denoted by bold lowercase (resp. uppercase) letters. For a distribution  $D$ , we write  $z \leftarrow D$  when the random variable  $z$  is sampled from  $D$ . For a real-valued function  $f$  and a countable set  $S$ , we write  $f(S) = \sum_{x \in S} f(x)$  assuming this sum is absolutely convergent. Let  $\ln$  denote the natural logarithm.

### 2.1 Linear Algebra and Lattices

Let  $b_i$  (resp.  $\mathbf{b}_i$ ) denote the  $i$ -th coordinate (resp. column) of  $\mathbf{b}$  (resp.  $\mathbf{B}$ ). Given  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ , their inner product is  $\langle \mathbf{u}, \mathbf{v} \rangle = \sum_{i=0}^{n-1} u_i v_i$ . Let  $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$  be the  $\ell_2$ -norm of  $\mathbf{v}$ .

Let  $\mathbf{B} = [\mathbf{b}_0, \dots, \mathbf{b}_{n-1}] \in \mathbb{R}^{m \times n}$  be a matrix of rank  $n$ . The Gram-Schmidt Orthogonalization (GSO) of  $\mathbf{B}$  is the unique matrix  $\tilde{\mathbf{B}} = [\tilde{\mathbf{b}}_0, \dots, \tilde{\mathbf{b}}_{n-1}] \in \mathbb{R}^{m \times n}$  such that  $\mathbf{B} = \tilde{\mathbf{B}}\mathbf{U}$  where  $\tilde{\mathbf{b}}_i$ 's are pairwise orthogonal and  $\mathbf{U}$  is upper-triangular with 1 on its diagonal. Let  $\|\mathbf{B}\|_{GS} = \max_i \|\tilde{\mathbf{b}}_i\|$ .

A lattice  $\Lambda$  generated by a basis  $\mathbf{B}$  is  $\Lambda = \Lambda(\mathbf{B}) = \{\mathbf{B}\mathbf{z} \mid \mathbf{z} \in \mathbb{Z}^n\}$ . The dual lattice of  $\Lambda$  is  $\Lambda^* = \{\mathbf{w} \in \text{span}(\Lambda) \mid \langle \mathbf{x}, \mathbf{w} \rangle \in \mathbb{Z}, \forall \mathbf{x} \in \Lambda\}$ . Many lattice-based schemes use a specific family of lattices known as  $q$ -ary lattice. Given  $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$ ,

let  $\Lambda_q^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{A}\mathbf{x} = \mathbf{0} \bmod q\}$ . Each  $\mathbf{u} \in \mathbb{Z}_q^n$  defines a lattice coset  $\Lambda_{\mathbf{u}}^\perp(\mathbf{A}) = \{\mathbf{x} \in \mathbb{Z}^m \mid \mathbf{A}\mathbf{x} = \mathbf{u} \bmod q\}$ .

## 2.2 NTRU

Let  $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$  with  $n = 2^k$  be a power-of-2 cyclotomic ring and  $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$  for a modulus  $q \in \mathbb{N}$ . For  $f = \sum_{i=0}^{n-1} f_i X^i \in \mathcal{R}$ , its  $\ell_2$ -norm is  $\|f\| = \sqrt{\sum_i f_i^2}$ . Similarly, we write  $\|(f, g)\| = \sqrt{\|f\|^2 + \|g\|^2}$  for  $f, g \in \mathcal{R}$ .

A NTRU lattice is a free module of rank 2 over  $\mathcal{R}$  defined by  $h \in \mathcal{R}_q$  as

$$\Lambda_{h,q} = \{(u, v) \in \mathcal{R}^2 \mid u + vh = 0 \bmod q\}.$$

In a typical NTRU-based scheme, the public key is  $h = f^{-1}g \bmod q$  for short  $f, g \in \mathcal{R}$  while  $(f, g)$  is used as the secret key. Two NTRU bases are of special interest:

$$\mathbf{B}_h = \begin{bmatrix} -h & q \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad \mathbf{B}_{f,g} = \begin{bmatrix} g & G \\ -f & -F \end{bmatrix}$$

where  $G, F \in \mathcal{R}$  such that  $fG - gF = q$  and  $\mathbf{B}_{f,g}$  is also called the NTRU trapdoor basis. In particular, FALCON uses a high-quality NTRU trapdoor such that  $\|B_{f,g}\|_{GS} \leq 1.17\sqrt{q}$ .

## 2.3 Gaussians

Let  $\rho_s(\mathbf{x}) = \exp(-\pi \cdot \|\mathbf{x}\|^2/s^2)$  with  $\mathbf{x} \in \mathbb{R}^n$  be the Gaussian function of width  $s$ . The width  $s$  is actually the scaled standard deviation of the corresponding Gaussian, i.e.  $s = \sigma\sqrt{2\pi}$ , where  $\sigma$  represents the standard deviation. We omit the subscript  $s$  if  $s = 1$ . Let  $\rho_{s,\mathbf{c}}(\mathbf{x}) = \rho_s(\mathbf{x} - \mathbf{c})$  for a center  $\mathbf{c} \in \mathbb{R}^n$ .

The *discrete Gaussian* on a lattice  $\Lambda$  with center  $\mathbf{c}$  and width  $s$  is defined by the probability function

$$D_{\Lambda,s,\mathbf{c}}(\mathbf{x}) = \frac{\rho_{s,\mathbf{c}}(\mathbf{x})}{\rho_{s,\mathbf{c}}(\Lambda)} \propto \rho_{s,\mathbf{c}}(\mathbf{x})$$

for  $\mathbf{x} \in \Lambda$ . The discrete Gaussian on  $\Lambda + \mathbf{c}$ , for  $\mathbf{c} \in \text{span}(\Lambda)$ , is defined by

$$D_{\Lambda+\mathbf{c},s}(\mathbf{x}) = \frac{\rho_s(\mathbf{x})}{\rho_s(\Lambda + \mathbf{c})}$$

for  $\mathbf{x} \in \Lambda + \mathbf{c}$ . Let  $\eta_\epsilon(\Lambda) = \min\{s > 0 \mid \rho_{1/s}(\Lambda^*) \leq 1 + \epsilon\}$  be the *smoothing parameter* of the lattice  $\Lambda$  with respect to the closeness parameter  $\epsilon \in (0, 1)$ . We write  $\bar{\eta}_\epsilon(\mathbb{Z}^m) = \sqrt{\ln(2m(1 + 1/\epsilon))/\pi}$  that is a tight upper bound of  $\eta_\epsilon(\mathbb{Z}^m)$ .

**Lemma 1** ([MR07], implicit in Lemma 4.4). *For any lattice  $\Lambda$ ,  $\mathbf{c} \in \text{span}(\Lambda)$  and  $s \geq \eta_\epsilon(\Lambda)$ , it holds that*

$$\rho_{s,\mathbf{c}}(\Lambda) \in \left[\frac{1-\epsilon}{1+\epsilon}, 1\right] \cdot \rho_s(\Lambda).$$

**Lemma 2** ([GPV08]). *Let  $\Lambda$  be an  $m$ -dimensional lattice with a basis  $\mathbf{B}$ , then*

$$\eta_\epsilon(\Lambda) \leq \|\mathbf{B}\|_{GS} \cdot \bar{\eta}_\epsilon(\mathbb{Z}^m).$$

## 2.4 Rényi Divergence

**Definition 1 (Rényi divergence).** Let  $P, Q$  be two distributions such that  $\text{Supp}(P) \subseteq \text{Supp}(Q)$ . For  $a \in (1, +\infty)$ , the Rényi divergence of order  $a$  is

$$R_a(P \parallel Q) = \left( \sum_{x \in \text{Supp}(P)} \frac{P(x)^a}{Q(x)^{a-1}} \right)^{\frac{1}{a-1}}.$$

In particular, the Rényi divergence of order  $+\infty$  is

$$R_\infty(P \parallel Q) = \max_{x \in \text{Supp}(P)} \frac{P(x)}{Q(x)}.$$

The Rényi divergence is a closeness measure between distributions. It leads to tighter bounds in many typical cryptographic usecases compared to the statistical distance-based arguments, as verified in [BLL<sup>+</sup>15, Pre17].

**Lemma 3 ([BLL<sup>+</sup>15, Pre17]).** For two distributions  $P, Q$ , two families of distributions  $(P_i)_i, (Q_i)_i$  and any function  $f$ , the Rényi divergence verifies the following properties:

- **Data processing inequality.**  $R_a(P^f \parallel Q^f) \leq R_a(P \parallel Q)$ .
- **Multiplicativity.**  $R_a(\prod_i P_i \parallel \prod_i Q_i) = \prod_i R_a(P_i \parallel Q_i)$ .
- **Probability preservation.** For any event  $E \subseteq \text{Supp}(Q)$  and  $a \in (1, +\infty)$ ,

$$Q(E) \geq P(E)^{a/(a-1)} / R_a(P \parallel Q),$$

$$Q(E) \geq P(E) / R_\infty(P \parallel Q).$$

**Lemma 4 (Optimal Rényi Order [FGdG<sup>+</sup>25]).** For  $\lambda \in \mathbb{N}$ , let  $\mathcal{E}_1, \mathcal{E}_2$  be two events such that  $\Pr[\mathcal{E}_1] \geq 2^{-\lambda}$ . Assume that for any  $Q \in \mathbb{N}$  and  $a \in (1, \infty)$  the Rényi divergence between two arbitrary distributions is at most  $R_a \in [1, \infty)$ , and

$$\Pr[\mathcal{E}_2] \leq R_a^Q \cdot \Pr[\mathcal{E}_1]^{\frac{a-1}{a}} \quad \forall a > 1.$$

Then

$$-\log(\Pr[\mathcal{E}_2]) \leq -\log(\Pr[\mathcal{E}_1]) - \min_{a>1} \left\{ Q \log R_a + \frac{\lambda}{a} \right\}.$$

**Definition 2 (Relative Error [MW17]).** Let  $\mathcal{P}$  and  $\mathcal{Q}$  be two discrete probability distributions over the same countable set  $\mathcal{X}$ . The relative error of  $\mathcal{P}$  and  $\mathcal{Q}$  is defined as

$$\delta_{RE}(\mathcal{P}, \mathcal{Q}) = \max_{x \in \text{sup}(\mathcal{P})} \frac{|\mathcal{P}(x) - \mathcal{Q}(x)|}{\mathcal{P}(x)}.$$

**Lemma 5 (Relative Lemma [Pre17]).** Let  $\mathcal{P}$  and  $\mathcal{Q}$  be two discrete probability distributions such that  $\text{sup}(\mathcal{P}) = \text{sup}(\mathcal{Q})$  and  $\delta_{RE} > 0$ . Then for all  $a \in (1, +\infty)$

$$R_a(\mathcal{P} \parallel \mathcal{Q}) \lesssim 1 + \frac{a\delta_{RE}^2}{2}.$$

### 3 FALCON-type Signatures

This paper mainly investigates the concrete security analysis of  $\text{FALCON}^+$  introduced by [FGdG<sup>+</sup>25] that is a minor modification of FALCON with a formal proof and  $\text{FALCON}^{\text{ws}}$  introduced by [ZJR<sup>+</sup>25] that is a recent variant of FALCON achieving better compactness. Let us first briefly recall  $\text{FALCON}^+$  and  $\text{FALCON}^{\text{ws}}$ .

#### 3.1 $\text{FALCON}^+$

A FALCON-type signature scheme is a NTRU-based instantiation of the GPV hash-and-sign framework [GPV08] over  $\mathcal{R} = \mathbb{Z}[X]/(X^n + 1)$  with  $n = 2^k$ . Its secret key  $\text{sk}$ , generated by  $\text{TpGen}$ , is a NTRU trapdoor basis  $\mathbf{B}_{f,g}$  of well-bounded Gram-Schmidt norms. The signing procedure is in essence preimage Gaussian sampling, denoted  $\text{PreSmp}$ , over some coset of  $\Lambda_{h,q}$  determined by the hash of message, salt and public key. A valid signature is a short Gaussian vector in the coset. Therefore the scheme is specified by the following parameters.

- the basis quality parameter  $\alpha > 1$  such that  $\|\mathbf{B}_{f,g}\|_{GS} \leq \alpha\sqrt{q}$ ;
- the Gaussian width  $s = \alpha\sqrt{q} \cdot \overline{\eta}_\epsilon(\mathbb{Z}^{2n})$  and the corresponding standard deviation  $\sigma = s/\sqrt{2\pi}$ ;
- the signature norm bound  $\beta = \tau \cdot \sigma\sqrt{2n}$  for some  $\tau \approx 1$ .

$\text{FALCON}^+$  is the first FALCON-type signature scheme with provable security. Compared to FALCON,  $\text{FALCON}^+$  picks a random salt refreshing the lattice coset within the loop of Gaussian sampling. With this minor modification, the reduction in the security proof can program the RO with a fresh, potentially “large” preimage without being forced to use a conditional distribution. The signing and verification procedure is described on a high level as follows:

| $\text{Sign}^+(\text{sk}, m)$                                     | $\text{Verify}(\text{pk}, m, \text{sig})$             |
|-------------------------------------------------------------------|-------------------------------------------------------|
| 1 : <b>repeat</b>                                                 | 1 : $c = \text{H}(\text{pk}, r, m) \in \mathcal{R}_n$ |
| 2 : $r \leftarrow \{0, 1\}^k$                                     | 2 : $s_1 = c - s_2 h \bmod q$                         |
| 3 : $c = \text{H}(\text{pk} = h, r, m)$                           | 3 : <b>if</b> $\ (s_1, s_2)\  \leq \beta$             |
| 4 : $(s_1, s_2) \leftarrow \text{PreSmp}^+(\text{sk}, s, (c, 0))$ | 4 : <b>accept</b>                                     |
| 5 : <b>until</b> $\ (s_1, s_2)\  \leq \beta$                      | 5 : <b>else</b>                                       |
| 6 : <b>return</b> $(\text{sig} = (r, s_2))$                       | 6 : <b>reject</b>                                     |

$\text{FALCON}^+$  uses the same parameters as FALCON:  $n = 512$  (resp. 1024) for NIST security level I (resp. V) and the modulus  $q = 12289$ . The basis quality parameter  $\alpha = 1.17$  and the signature norm bound  $\beta = \tau\sigma\sqrt{2n}$  with a fixed tailcut factor  $\tau = 1.1$ . The closeness parameter is set according to the Rényi divergence-based argument [Pre17] as  $\epsilon = 1/\sqrt{Q_s \cdot \lambda}$ , where  $Q_s = 2^{64}$  is the



maximal number of signing queries and  $\lambda = 128$  or  $256$  is the security parameter. The preimage sampler  $\text{PreSmp}^+$  is implemented with the FFO sampler [DP16] that is a ring-efficient Klein-GPV sampler.

The provable security of  $\text{FALCON}^+$  is based on the following hardness assumptions.

**Definition 3** ( $\mathcal{R}\text{-SIS}$ ,  $t\text{-}\mathcal{R}\text{-ISIS}$ ). *The Ring Short Integer Solution problem and the Ring Inhomogeneous Short Integer Solution problem relative to the NTRU trapdoor algorithm  $\text{TpdGen}$  with parameters  $m, q > 0$  and  $\alpha, B > 0$  are defined via the games  $\mathcal{R}\text{-SIS}$  and  $t\text{-}\mathcal{R}\text{-ISIS}$  in Fig. 1. The advantages of  $\mathbf{A}$  is defined as*

$$\begin{aligned}\text{Adv}_{m,q,\alpha,B,\mathbf{A}}^{\mathcal{R}\text{-SIS}} &= \Pr[\mathcal{R}\text{-SIS}_{m,q,\alpha,B}(\mathbf{A}) \Rightarrow 1], \\ \text{Adv}_{m,q,\alpha,B,\mathbf{A}}^{t\text{-}\mathcal{R}\text{-ISIS}} &= \Pr[t\text{-}\mathcal{R}\text{-ISIS}_{m,q,\alpha,B}(\mathbf{A}) \Rightarrow 1].\end{aligned}$$

|                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Game <math>\mathcal{R}\text{-SIS}_{m,q,\alpha,B}(\mathbf{A})</math></b>                                                                                                                                                                                                                                                                                                                      |
| 1 : <b>for</b> $i \in [m]$ <b>do</b><br>2 : $(h_i, \cdot, \cdot) \leftarrow \text{TpdGen}(q, \alpha)$<br>3 : $(u, v_1, \dots, v_m) \leftarrow \mathbf{A}(h_1, \dots, h_m)$<br>4 : <b>return</b> $\llbracket u + \sum_i h_i \cdot v_i = 0 \wedge \ (u, v_1, \dots, v_m)\  \leq B \rrbracket$                                                                                                     |
| <b>Game <math>t\text{-}\mathcal{R}\text{-ISIS}_{m,q,\alpha,B}(\mathbf{A})</math></b>                                                                                                                                                                                                                                                                                                            |
| 1 : <b>for</b> $i \in [m]$ <b>do</b><br>2 : $(h_i, \cdot, \cdot) \leftarrow \text{TpdGen}(q, \alpha)$<br>3 : <b>for</b> $j \in [t]$ <b>do</b><br>4 : $c_j \leftarrow \mathcal{R}_q$<br>5 : $(j, u, v_1, \dots, v_m) \leftarrow \mathbf{A}(h_1, \dots, h_m, c_1, \dots, c_t)$<br>6 : <b>return</b> $\llbracket u + \sum_i h_i \cdot v_i = c_j \wedge \ (u, v_1, \dots, v_m)\  \leq B \rrbracket$ |

**Fig. 1.** Games defining  $\mathcal{R}\text{-SIS}_{m,q,\alpha,B}$  and  $t\text{-}\mathcal{R}\text{-ISIS}_{m,q,\alpha,B}$

By Rényi divergence-based arguments, the security proofs in the Random Oracle Model (ROM) of  $\text{FALCON}^+$  are presented in [FGdG<sup>+</sup>25]: the strong unforgeability is reduced to  $\mathcal{R}\text{-SIS}$  while the plain unforgeability is reduced to  $t\text{-}\mathcal{R}\text{-ISIS}$ . Theorem 1 offers a characterization of the plain unforgeability of  $\text{FALCON}^+$  and a detailed account of core issues we are concerned.

**Theorem 1** (Unforgeability, [FGdG<sup>+</sup>25]). *For any adversary  $\mathbf{A}$  against the UF-CMA security of  $\text{COREFALCON}^+$  running in time  $t_{\mathbf{A}}$ , making at most  $Q_s$  signing queries and  $Q_{\text{H}}$  RO queries, there exist adversary  $\mathbf{B}$  against  $(Q_{\text{H}} + 1)\text{-}\mathcal{R}\text{-ISIS}$  running in time  $t_{\mathbf{B}} \approx t_{\mathbf{A}}$  such that for all  $C_s \in \mathbb{N}^{\geq 1}$  and  $a_u, a_p \in \mathbb{R}^{>1}$  it holds*

$$\begin{aligned} \text{Adv}_{\text{FALCON}^+, \mathbf{A}}^{Q_s\text{-UF-CMA}} &\leq \left( r_u^{C_s} \cdot \left( r_p^{C_s} \cdot \text{Adv}_{1,q,\alpha,\beta,\mathbf{B}}^{(Q_H+1)\text{-}\mathcal{R}\text{-ISIS}} \right)^{\frac{a_p-1}{a_p}} \right)^{\frac{a_u-1}{a_u}} \\ &\quad + \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp}^+, \beta})^{C_s-i} (p_{\text{PreSmp}^+, \beta})^i + \frac{Q_s(C_s + Q_H)}{2^k}, \end{aligned}$$

where

$$\begin{aligned} p_{\text{PreSmp}^+, \beta} &= \max_{(\mathbf{B}, \cdot) \in \text{TpdGen}, c \in \mathcal{R}_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}^+(\mathbf{B}, s, (c, 0))} [\| (s_1, s_2) \|_2 \leq \beta], \\ r_u &= \max_{(\cdot, h) \in \text{TpdGen}} R_{a_u}(\mathcal{P} \parallel \mathcal{Q}_h) \text{ with } \mathcal{P} = \mathcal{U}(\mathcal{R}_q) \text{ and } \mathcal{Q}_h \text{ the distribution of} \\ &\quad s_1 + s_2 \cdot h \bmod q, \text{ where } s_1, s_2 \sim D_{\mathcal{R}, s}, \\ r_p &= \max_{(\mathbf{B}, \cdot) \in \text{TpdGen}, c \in \mathcal{R}_q} R_{a_p}(\text{PreSmp}^+(\mathbf{B}, s, (c, 0)) \parallel \mathcal{D}_{\Lambda(\mathbf{B}), s, (c, 0)}). \end{aligned}$$

**Security reduction losses in  $\text{FALCON}^+$ .** In Theorem 1,  $\text{Adv}_{1,q,\alpha,\beta,\mathbf{B}}^{(Q_H+1)\text{-}\mathcal{R}\text{-ISIS}}$  is the advantage against the underlying ISIS problem, and  $r_p$  and  $r_u$  are the Rényi divergences with respect to the underlying sampler and RO simulation, respectively. Note that the first term in  $\text{Adv}_{1,q,\alpha,\beta,\mathbf{B}}^{(Q_H+1)\text{-}\mathcal{R}\text{-ISIS}}$  indicates that there are two sources of security losses in the concrete security of  $\text{FALCON}^+$ .

1. *The loss from preimage sampler:* The actual preimage sampler  $\text{PreSmp}^+$  usually does not output a perfect Gaussian  $D_{\Lambda(\mathbf{B}), s, (c, 0)}$ . The Rényi divergence between the real preimage distribution and the ideal one is bounded by  $r_p = R_{a_p}(\text{PreSmp}^+ \parallel D_{\Lambda(\mathbf{B}), s, (c, 0)}) \leq 1 + 2a_p\epsilon^2$ . The term  $r_p$  is exponentiated by  $C_s$ , the maximal number of the calls of  $\text{PreSmp}^+$  (including repetitions). Then the multiplicative factor  $r_p^{C_s}$  of  $\text{Adv}_{1,q,\alpha,\beta,\mathbf{B}}^{(Q_H+1)\text{-}\mathcal{R}\text{-ISIS}}$  indicates that the sampler incurs  $l_p = C_s \cdot \log_2(r_p)$  bits of security loss.
2. *The loss from RO Simulation:* In the security proof, the simulator must answer RO queries  $\mathbf{H}(\mathbf{pk}, \mathbf{r}, \mathbf{m})$  without knowing the trapdoor. It does so by programming the RO output as  $c = s_1 + s_2 h \bmod q$  with  $s_1, s_2 \leftarrow D_{\mathcal{R}, s}$ . The distribution  $\mathcal{Q}_h$  of these programmed outputs is not perfectly uniform over  $\mathcal{R}_q$ . The term  $r_u = R_{a_u}(\mathcal{U}(\mathcal{R}_q) \parallel \mathcal{Q}_h) \approx 1 + \frac{2a_u\epsilon^2}{(1-\epsilon)^2}$  quantifies the Rényi divergence between the ideal uniform distribution and this simulated distribution. It is exponentiated by  $C_s$  as well, thus the RO simulation incurs  $l_u = C_s \cdot \log_2(r_u)$  bits of security loss.

As claimed in [FGdG+25], the total security loss in the plain unforgeability analysis from two Rényi arguments was approximately 7 bits for  $\text{FALCON}^+$ -512 and 8 bits for  $\text{FALCON}^+$ -1024. A closely matching loss profile was observed in the strong unforgeability setting, although the underlying SIS problem becomes substantially easier in the security proof for strong unforgeability.

### 3.2 FALCON<sup>ws</sup>

To further improve the bandwidth efficiency, the recent work [ZJR<sup>+</sup>25] introduces a FALCON variant FALCON<sup>ws</sup> working with the so-called weak smoothness condition, i.e.  $\epsilon = O(1)$ . This leads to a much smaller Gaussian width  $s = \alpha\sqrt{q} \cdot \eta_e(\mathbb{Z}^{2n})$ , which allows smaller parameters and higher forgery security.

The preimage distribution output by the original Klein-GPV sampler will have a significant discrepancy from the ideal Gaussian under the weak smoothness condition. To address this issue, [ZJR<sup>+</sup>25] exploits the rejection sampling to rectify the sampling error (Step 5 in Fig. 2). By this, the preimage in FALCON<sup>ws</sup> exactly follows the ideal Gaussian, which actually eliminates the Rényi loss from the sampler in the security proof.

| PreSmp <sup>ws</sup> (sk, s, (c, 0)) with weak smoothness |                                                                                                                                                |
|-----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 :                                                       | $\mathbf{v} \leftarrow \mathbf{0}, \mathbf{c}' \leftarrow \mathbf{c}, \Delta \leftarrow 1$                                                     |
| 2 :                                                       | <b>for</b> $i = n - 1, \dots, 0$                                                                                                               |
| 3 :                                                       | $c_i'' \leftarrow \langle \mathbf{c}', \tilde{\mathbf{b}}_i \rangle / \ \tilde{\mathbf{b}}_i\ ^2, s_i \leftarrow s / \ \tilde{\mathbf{b}}_i\ $ |
| 4 :                                                       | $z_i \leftarrow D_{\mathbb{Z}, s_i, c_i''}$                                                                                                    |
| 5 :                                                       | $\Delta \leftarrow \Delta \cdot \frac{\rho_{s_i, c_i''}(\mathbb{Z})}{\rho_{s_i}(\mathbb{Z})}$                                                  |
| 6 :                                                       | $\mathbf{c}' \leftarrow \mathbf{c}' - z_i \mathbf{b}_i, \mathbf{v} \leftarrow \mathbf{v} + z_i \mathbf{b}_i$                                   |
| 7 :                                                       | <b>return</b> $\mathbf{v}$ with probability $\Delta$ otherwise restart                                                                         |

**Fig. 2.** Preimage sampler used in FALCON<sup>ws</sup>.

However, as noted before, there is another source of reduction loss from the RO simulation. The use of an excessively large  $\epsilon$  in FALCON<sup>ws</sup> would drastically amplify this loss, which invalidates the security reduction. To build a theoretical security proof, [ZJR<sup>+</sup>25] introduces an additional non-standard hardness assumption to guarantee computational indistinguishability of the simulated syndrome from uniform. While they gave some cryptanalytic results to show that the new assumption does not degrade concrete security, removing such an extra assumption would be much more reassuring.

## 4 Sharper Security Bounds Using the NTRU Basis Profile

In this section, we leverage the profile of the trapdoor basis to improve the Rényi divergence-based security arguments for the GPV framework. Specifically, we present *basis-specific* Rényi divergence bounds for both the Klein-GPV sampler [GPV08] and the RO simulation in the GPV security proof. By applying

the new analysis to FALCON<sup>+</sup>, we can eliminate the 7 bits of security loss for the FALCON<sup>+</sup>-512 parameter set claimed in [FGdG<sup>+</sup>25].

#### 4.1 Basis-Specific Rényi Analysis of the Klein-GPV Sampler

The closeness error between the output distribution by the trapdoor sampler and the ideal Gaussian is a primary source of the security loss in the GPV framework. The Klein-GPV sampler, adopted by FALCON, reduces the high-dimensional lattice Gaussian sampling to a sequence of one-dimensional integer Gaussian samplings with varying parameters. The Gaussian width at each iteration is  $s_i = s/\|\tilde{\mathbf{b}}_i\|$  inversely proportional to the Gram-Schmidt norm of the NTRU trapdoor. The Klein-GPV sampler requires all  $s_i$ 's are greater than  $\eta_\epsilon(\mathbb{Z}) \approx \sqrt{\ln(2(1+1/\epsilon))}$  for a small closeness parameter  $\epsilon$ , which gives a Rényi divergence bound for the signature distribution and Gaussian.

The existing Rényi analysis is based on the almost shift invariant property of integer Gaussians with width greater than the smoothing parameter. Let  $\epsilon_i > 0$  such that  $s_i = \sqrt{\ln(2(1+1/\epsilon_i))/\pi} \geq \eta_{\epsilon_i}(\mathbb{Z})$ . We observe that  $\epsilon_i$  is highly sensitive to  $\|\tilde{\mathbf{b}}_i\|$ :  $\epsilon_i$ , i.e. the sampling error at the  $i$ -th iteration, can be significantly smaller than the global  $\epsilon$  with  $\|\tilde{\mathbf{b}}_i\|$  decreasing. The Rényi divergence bound can therefore be improved by taking all  $s_i$ 's into account. To this end, we present a refined Rényi analysis using a basis-specific metric

$$\delta_{\mathbf{B},s} = \prod_{i=1}^m \frac{1+\epsilon_i}{1-\epsilon_i} - 1.$$

**Theorem 2 (Basis-Specific Relative Error of the Klein-GPV Sampler).**

For any basis  $\mathbf{B} \in \mathbb{Z}^{m \times m}$ ,  $s \geq \|\mathbf{B}\|_{GS} \cdot \eta_\epsilon(\mathbb{Z}^m)$  with  $\epsilon \in (0,1)$  and any target  $\mathbf{t} \in \mathbb{Z}^m$ , the relative error of the Klein-GPV Sampler  $\text{PreSmp}(\mathbf{B}, s, \mathbf{t})$  is bounded by

$$\delta_{RE}(\text{PreSmp}(\mathbf{B}, s, \mathbf{t}), D_{\mathcal{L}(\mathbf{B}),s,\mathbf{t}}) \leq \delta_{\mathbf{B},s}.$$

*Proof.* As detailed in [GPV08], the probability that the Klein-GPV Sampler  $\text{PreSmp}(\mathbf{B}, s, \mathbf{t})$  outputs a vector  $\mathbf{v}$  is proportional to

$$\prod_{i=1}^m \frac{1}{\rho_{s_i, c_i}(\mathbb{Z})} \cdot \rho_{s, \mathbf{c}}(\mathbf{v}),$$

where  $s_i = s/\|\tilde{\mathbf{b}}_i\|$  and some real  $c_i$ . Since each  $s_i \geq \eta_{\epsilon_i}(\mathbb{Z})$  for some  $\epsilon_i$ , Lemma 1 shows that  $\rho_{s_i, c_i}(\mathbb{Z}) \in [\frac{1-\epsilon_i}{1+\epsilon_i}, 1] \cdot \rho_{s_i}(\mathbb{Z})$ . The claimed bound follows immediately.  $\square$

**Corollary 1 (Basis-Specific Rényi Divergence of the Klein-GPV Sampler).** Under the same conditions as Theorem 2, for any  $a > 1$ , the Rényi divergence is bounded by

$$R_a(\text{PreSmp}(\mathbf{B}, s, \mathbf{t}), D_{\mathcal{L}(\mathbf{B}),s,\mathbf{t}}) \lesssim 1 + \frac{a\delta_{\mathbf{B},s}^2}{2}.$$

*Remark 1.* The key advancement in our analysis is its basis-specific nature. Instead of using the worst-case lower bound as in [Pre17], our results are tailored to *each specific basis*  $\mathbf{B}$ . This enables tighter bounds in practice, as we leverage the actual geometric properties of  $\mathbf{B}$  rather than conservative worst-case estimates. Our experimental evaluation validates the efficacy of this analytical approach in achieving a tight security reduction.

## 4.2 Basis-Specific Rényi Analysis of the RO Simulation

This subsection addresses the Rényi loss from the RO simulation through a refined analysis of the smoothing parameter that accounts for the geometric profile of the lattice basis. Building on this tighter bound and a more detailed treatment of the relative error in the original analysis of [FGdG<sup>+</sup>25], we obtain significantly improved bounds for the Rényi loss in the RO simulation.

**Theorem 3.** *Let  $\mathbf{B} \in \mathbb{Z}^{m \times m}$  be a basis of the lattice  $\Lambda$  and  $\alpha_i = \frac{\|\mathbf{B}\|_{GS}}{\|\mathbf{b}_i\|}$ . Let  $s = \|\mathbf{B}\|_{GS} \cdot \sqrt{\ln(2m(1+1/\epsilon))}/\pi$ , then*

$$\rho_{1/s}(\Lambda^* \setminus \mathbf{0}) \leq \frac{\sum_{i=0}^{m-1} 2 \left( \frac{\epsilon}{2m(1+\epsilon)} \right)^{\alpha_i^2}}{1 - \sum_{i=0}^{m-1} 2 \left( \frac{\epsilon}{2m(1+\epsilon)} \right)^{\alpha_i^2}}.$$

Defining the right side of the inequality as  $\epsilon'$ , then  $\eta_{\epsilon'}(\Lambda) \leq \|\mathbf{B}\|_{GS} \cdot \sqrt{\ln(2m(1+1/\epsilon))}/\pi$ .

*Proof.* Note that for the dual lattice  $\Lambda^*$ , the reverse order GSO of dual basis  $\tilde{\mathbf{B}}^*$  satisfies  $\tilde{\mathbf{b}}_i^* = \frac{\tilde{\mathbf{b}}_i}{\|\mathbf{b}_i\|^2}$ . Then let us consider the fundamental parallelepiped (scaled)

$$\mathcal{P}(\tilde{\mathbf{B}}^*) = \{\tilde{\mathbf{B}}^* \mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n, \forall i : -1 < x_i < 1\}.$$

Obviously, there are no other lattice points of  $\Lambda^*$  inside  $\mathcal{P}(\tilde{\mathbf{B}}^*)$  except the zero point  $\mathbf{0}$ . Denote  $\mathcal{Q}_i = \left\{ \mathbf{x} \in \mathbb{R}^m \mid \left| \langle \mathbf{x}, \frac{\tilde{\mathbf{b}}_i^*}{\|\tilde{\mathbf{b}}_i^*\|} \rangle \right| \geq \|\tilde{\mathbf{b}}_i^*\| \right\}$ . It follows from Lemma 2.4 in [Ban95] that

$$\rho_{1/s}(\Lambda^* \cap \mathcal{Q}_i) < 2e^{-\pi s^2 \|\tilde{\mathbf{b}}_i^*\|^2} \rho_{1/s}(\Lambda^*).$$

Notice that

$$\Lambda^* \setminus \mathcal{P}(\tilde{\mathbf{B}}^*) = \Lambda^* \cap (\mathcal{Q}_0 \cup \dots \cup \mathcal{Q}_{m-1}) = \cup_{i=0}^{m-1} (\Lambda^* \cap \mathcal{Q}_i).$$

then it follows that

$$\begin{aligned}
\rho_{1/s}(\Lambda^* \setminus \mathbf{0}) &= \rho_s(\Lambda^* \setminus \mathcal{P}(\tilde{\mathbf{B}}^*)) \leq \sum_{i=0}^{m-1} \rho_s(\Lambda^* \cap \mathcal{Q}_i) \\
&< 2\rho_{1/s}(\Lambda^*) \sum_{i=0}^{m-1} e^{-\pi s^2 \|\tilde{\mathbf{b}}_i^*\|^2} \\
&= 2\rho_{1/s}(\Lambda^*) \sum_{i=0}^{m-1} e^{-\pi \left( \|\mathbf{B}\|_{GS} \cdot \sqrt{\ln(2m(1+\epsilon))/\pi} \right)^2 \cdot \frac{1}{\|\mathbf{b}_i^*\|^2}} \\
&= \sum_{i=0}^{m-1} 2 \left( \frac{\epsilon}{2m(1+\epsilon)} \right)^{\alpha_i^2} \cdot \rho_{1/s}(\Lambda^*).
\end{aligned}$$

Rearranging terms, we can get the upper bound estimate for  $\rho_{1/s}(\Lambda^* \setminus \mathbf{0})$ .  $\square$

*Remark 2.* In FALCON, the Gaussian width  $s$  for preimage sampling is set with respect to closeness parameter  $\epsilon$ . However, our basis-specific analysis reveals that it corresponds to a significantly smaller effective parameter  $\epsilon'$ . As shown later, the exponent of  $\epsilon'$  is approximately 5 bits smaller than that of the original  $\epsilon$ .

Next, we observe a potential for improvement in the original analysis of Lemma 8 in [FGdG<sup>+</sup>25], which bounds the relative error between the uniform distribution  $\mathcal{U}(\mathcal{R}_q)$  and the simulated distribution  $\mathcal{Q}_h$  by a factor derived from  $\frac{1+\epsilon}{1-\epsilon}$ . This value implicitly overlooks a critical subtlety: the  $\epsilon$  terms in the numerator and denominator actually correspond to two distinct closeness parameters with significantly different magnitudes.

**Theorem 4 (Refined Uniformity Bound, adapted from Lemma 8 of [FGdG<sup>+</sup>25]).** *Let  $\Lambda, \Lambda'$  be  $m$ -dimensional full-rank lattices with  $\Lambda' \subseteq \Lambda$ . Then for any  $a \in (1, \infty)$ ,  $\epsilon \in (0, \frac{1}{2})$ , any  $s \geq \eta_\epsilon(\Lambda')$ , and any  $\mathbf{c} \in \mathbb{R}^n$ ,*

$$R_a(\mathcal{U}(\Lambda/\Lambda') \| D_{\Lambda/\Lambda', s, \mathbf{c}}) \lesssim 1 + \frac{a\delta^2}{2},$$

where  $\delta = (\epsilon + \epsilon')/(1 - \epsilon)$ .

*Proof.* The quotient group  $\Lambda/\Lambda'$  is defined as the additive group of cosets  $\mathbf{x} + \Lambda'$ ,  $\mathbf{x} \in \Lambda$ . Sampling from a discrete Gaussian over this quotient group we obtain that for any  $\mathbf{x} \in \Lambda$

$$D_{\Lambda/\Lambda', s, \mathbf{c}}(x) = \frac{\rho_{s, \mathbf{c}}(\mathbf{x} + \Lambda')}{\rho_{s, \mathbf{c}}(\Lambda)}.$$

Notice that since  $s \geq \eta_\epsilon(\Lambda')$  and  $\Lambda' \subset \Lambda$ , there exists  $\epsilon' \ll \epsilon$  such that  $s \geq \eta_{\epsilon'}(\Lambda)$ . As a consequence, we get

$$\frac{\mathcal{U}(|\Lambda/\Lambda'|)(\mathbf{x})}{D_{\Lambda/\Lambda', s, \mathbf{c}}(\mathbf{x})} \in \left[ \frac{1-\epsilon}{1+\epsilon'}, \frac{1+\epsilon'}{1-\epsilon} \right].$$

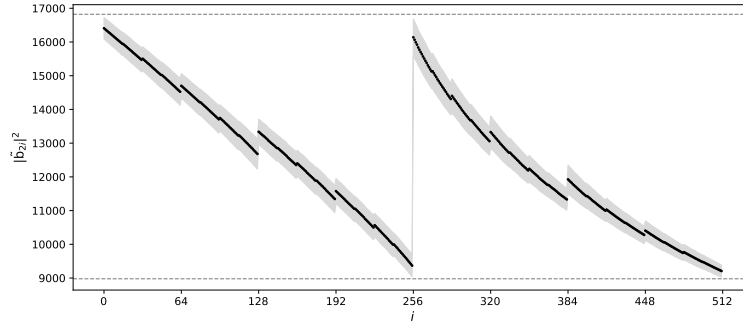
Applying the relative lemma 5 with  $\delta = (\epsilon + \epsilon')/(1 - \epsilon)$ , we complete the proof.  $\square$

*Remark 3.* In [FGdG+25], the relative error is bounded by  $2\epsilon/(1-\epsilon)$ , leading to a Rényi divergence upper bound of  $1 + 2a\epsilon^2/(1-\epsilon)^2$ . In contrast, our analysis employs a tighter bound of  $\delta = (\epsilon + \epsilon')/(1-\epsilon)$ , where the term  $\epsilon'$  is negligible for the FALCON parameters (for example,  $\epsilon' = 2^{-782250}$ ). Consequently, the relative error in our work is effectively halved compared to theirs, which results in a Rényi divergence bound that is *approximately twice* as tight.

In summary, by combining a basis-specific analysis of the smoothing parameter with a more refined treatment of the relative error, our experimental results demonstrate that the resulting Rényi loss from the RO simulation becomes numerically comparable to that originating from the sampler. See Table 5 for more details.

### 4.3 Analysis and Experimental Results for FALCON<sup>+</sup>

**The GSO Profile of NTRU Trapdoor Bases.** It is known that  $\|\tilde{\mathbf{b}}_{2i}\| = \|\tilde{\mathbf{b}}_{2i+1}\|$  for each FALCON trapdoor basis. As per the symplecticity of NTRU [GHN06], it holds that  $\|\tilde{\mathbf{b}}_i\| * \|\tilde{\mathbf{b}}_{2n-i}\| = q$  and this indicates that  $\|\tilde{\mathbf{b}}_i\| \in [\sqrt{q}/1.17, 1.17\sqrt{q}]$ . Figure 3 shows the average profile of FALCON trapdoor bases.



**Fig. 3.** The black scatter points represent the mean of  $\|\tilde{\mathbf{b}}_{2i}\|^2$  for the 1,000 random FALCON-512 basis, and the gray area represents the standard deviation. The values of the two horizontal dashed lines are  $(1.17\sqrt{q})^2$  and  $(\sqrt{q}/1.17)^2$ .

**Rényi Divergence Estimation of the Klein-GPV Sampler.** Based on our improved analysis in Corollary 1 and the characterization of the NTRU basis profile, we present a conservative estimate of the optimization achieved. Note that in Theorem 2,

$$\begin{aligned} s_i &= s/\|\tilde{\mathbf{b}}_i\| = \alpha_i \cdot \sqrt{\ln(2(1+1/\epsilon))/\pi} \\ &= \sqrt{\ln(2(1+1/\epsilon))^{\alpha_i^2}/\pi} > \sqrt{\ln(2(1+1/\epsilon^{\alpha_i^2}))/\pi} \\ &\geq \eta_{\epsilon'_i}(\mathbb{Z}^n) \end{aligned}$$

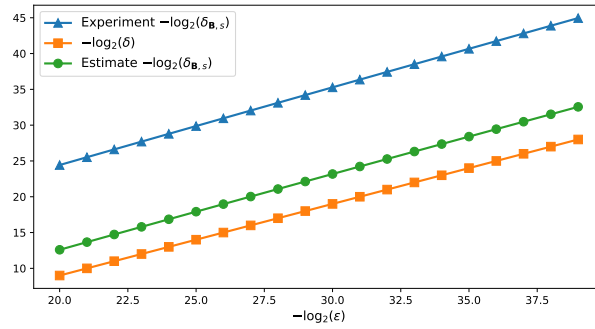
for  $\epsilon'_i = \epsilon^{\alpha_i^2}$ . As a result, the relative error between  $\text{PreSmp}(\mathbf{B}, s, \mathbf{t})$  and  $D_{\mathcal{L}(\mathbf{B}), s, \mathbf{t}}$  is bounded by

$$\delta_{\mathbf{B}, s} = \prod_{i=1}^{2n} \frac{1 + \epsilon'_i}{1 - \epsilon'_i} - 1 \approx 2 \sum_{i=1}^{2n} \epsilon'_i.$$

We derive a heuristic estimate via integral approximation as follows:

$$\begin{aligned} 2 \sum_{i=1}^{2n} \epsilon'_i &= 4 \sum_{i=0}^{n-1} \epsilon^{\alpha_{2i}^2} \\ &\approx \epsilon \cdot \frac{4n}{1.17^4 - 1} \int_0^{1.17^4 - 1} \epsilon^x dx \\ &= \frac{4n\epsilon}{1.17^4 - 1} \cdot \frac{1 - \epsilon^{1.17^4 - 1}}{|\ln \epsilon|} \\ &\approx \frac{4n\epsilon}{0.87 |\ln \epsilon|}. \end{aligned} \tag{1}$$

Recall that  $\delta_{\mathbf{B}, s} = \prod_{i=1}^m \frac{1 + \epsilon_i}{1 - \epsilon_i} - 1$  is a basis-specific metric. Compared to the original relative error  $\delta = \left(\frac{1 + \epsilon}{1 - \epsilon}\right)^{2n} - 1 \approx 4n\epsilon$ , we improve by a factor of  $\delta/\delta_{\mathbf{B}, s} \approx 0.87 |\ln \epsilon| \approx 21.4$  for FALCON, implying that the actual relative error caused by the sampling algorithm is about 4.4 bits lower than previously estimated. The experimental results are shown in Figure 4: The relative error obtained from tests on 1,000 sets of random FALCON bases is smaller than the estimate from Equation 1, which implies a smaller Rényi divergence.



**Fig. 4.** The experiment  $\delta_{\mathbf{B}, s}$  comes from the mean of  $\delta_{\mathbf{B}, s}$  for the 1,000 random FALCON-512 basis.

**Rényi Divergence Estimation of the RO Simulation.** With the knowledge of the NTRU trapdoor profile, one can use Theorem 3 to estimate  $\epsilon'$  in  $\eta_{\epsilon'}(\Lambda_{NTRU}) \leq 1.17\sqrt{q} \cdot \sqrt{\ln(4n(1 + 1/\epsilon))}/\pi$  that corresponds to the Gaussian



width  $s$  in FALCON. We also give a heuristic estimate using integral approximation as follows:

$$\begin{aligned}
\sum_{i=0}^{2n-1} 2 \left( \frac{\epsilon}{4n(1+\epsilon)} \right)^{\alpha_i^2} &= 4 \sum_{i=0}^{n-1} \left( \frac{\epsilon}{4n(1+\epsilon)} \right)^{\alpha_{2i}^2} \\
&\approx \frac{\epsilon}{n(1+\epsilon)} \cdot \frac{n}{1.17^4 - 1} \int_0^{1.17^4 - 1} \left( \frac{\epsilon}{4n(1+\epsilon)} \right)^x dx \\
&= \frac{\epsilon}{1+\epsilon} \cdot \frac{1}{1.17^4 - 1} \cdot \frac{1 - \left( \frac{\epsilon}{4n(1+\epsilon)} \right)^{1.17^4 - 1}}{\left| \ln \left( \frac{\epsilon}{4n(1+\epsilon)} \right) \right|} \\
&\approx \frac{1.15\epsilon}{|\ln(\epsilon/4n)|}.
\end{aligned} \tag{2}$$

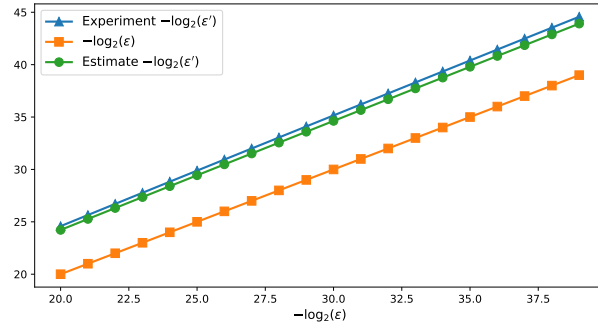
Then by Theorem 3, it follows that  $\epsilon'$  is about  $\frac{1.15}{|\ln(\epsilon/4n)|}$  times smaller than  $\epsilon$  in width  $s$ . As shown in Figure 5, the relative error obtained from testing 1,000 sets of random FALCON bases closely matches the estimate derived from Equation 4.3, validating the accuracy of our estimation method. We present the numbers associated with FALCON for a straightforward comparison:

– for FALCON-512:

$$\eta_{2-40.9}(\Lambda_{NTRU}) \leq 1.17\sqrt{q} \cdot \sqrt{\ln(4n(1 + 2^{35.5}))/\pi}$$

– for FALCON-1024:

$$\eta_{2-41.2}(\Lambda_{NTRU}) \leq 1.17\sqrt{q} \cdot \sqrt{\ln(4n(1 + 2^{36}))/\pi}$$



**Fig. 5.** The experiment  $\epsilon'$  comes from the mean of  $\epsilon'$  for the 1,000 random FALCON-512 basis. The estimate  $\epsilon'$  is set as  $\frac{1.15\epsilon}{|\ln(\epsilon/4n)|}$ .

**Experiments on FALCON Bases.** We conducted extensive experiments to validate the effectiveness of our basis-specific analysis. As shown in Table 5, our improved analysis achieves a Rényi loss of 0.077 bits in average and the maximal loss among 1,000 random FALCON instances is less than 0.2 bits.

**Table 5.** Comparisons of the bit losses (in bits) achieved by [FGdG<sup>+</sup>25] and our analysis. The item  $A/B$  represents the mean and the maximum measured over 1,000 FALCON instances.

|                           | $loss_p$               |                   | $loss_u$               |                   |
|---------------------------|------------------------|-------------------|------------------------|-------------------|
|                           | [FGdG <sup>+</sup> 25] | Ours              | [FGdG <sup>+</sup> 25] | Ours              |
| FALCON <sup>+</sup> -512  | 3.29                   | 0.03854 / 0.09312 | 3.14                   | 0.03852 / 0.09299 |
| FALCON <sup>+</sup> -1024 | 3.14                   | 0.04563 / 0.08663 | 3.46                   | 0.04561 / 0.08658 |

#### 4.4 Refined Concrete Security of FALCON<sup>+</sup>

Putting everything together, we get a refined concrete security assessment of FALCON<sup>+</sup>. The resulting Rényi losses under our basis-specific analysis are summarized in Table 6, along with the bounds from [FGdG<sup>+</sup>25] for comparison.

**Table 6.** Concrete security of FALCON<sup>+</sup> for plain unforgeability. For FALCON<sup>+</sup>-1024, the bit security estimates by both [FGdG<sup>+</sup>25] and ours are 256, which are actually determined by the term  $\frac{Q_s(C_s+Q_H)}{2^k}$  in Theorem 1. For this, we also include the bit security ignoring that term in the bracket.

| Parameter                                  | FALCON <sup>+</sup> -512 |         | FALCON <sup>+</sup> -1024 |              |
|--------------------------------------------|--------------------------|---------|---------------------------|--------------|
|                                            | [FGdG <sup>+</sup> 25]   | Ours    | [FGdG <sup>+</sup> 25]    | Ours         |
| Degree of $\mathcal{R}$                    | 512                      |         | 1024                      |              |
| Bit security for $t$ - $\mathcal{R}$ -ISIS | 120                      |         | 278                       |              |
| Max Signing queries, $Q_s$                 | $2^{64}$                 |         | $2^{64}$                  |              |
| Max repetitions, $C_s$                     | $2^{64} + 2^{50}$        |         | $2^{64} + 2^{36}$         |              |
| Rényi Order, $a_p$                         | 72.96                    | 2577.30 | 157.05                    | 6417.73      |
| Rényi Order, $a_u$                         | 69.64                    | 2573.91 | 153.28                    | 6413.92      |
| Bits lost from $r_p^{C_s}, loss_p$         | 3.29                     | 0.09312 | 3.54                      | 0.08663      |
| Bits lost from $r_u^{C_s}, loss_u$         | 3.14                     | 0.093   | 3.46                      | 0.08658      |
| Final bit security                         | 113.17                   | 119.81  | 256 (271)                 | 256 (277.82) |

*Remark 4.* Note that the security loss values for our analysis in Table 6 correspond to the maximum measured over 1,000 instances. This constitutes a conservative estimate, and we are therefore confident that for the majority of FALCON trapdoor bases, the actual reduction loss will not exceed the reported values. For a completely proven assurance, one can add some checks in the key generation to guarantee the used basis achieving the security bounds.

## 5 On the Provable Security of FALCON<sup>ws</sup>

Despite its better compactness and higher bit security, the lack of a formal proof based on standard assumptions is seen as a major limitation of FALCON<sup>ws</sup>. Driven by this, we investigate how large the closeness parameter  $\epsilon$  can be to allow for a formal proof relying exclusively on standard hardness assumptions while maintaining the advantage of FALCON<sup>ws</sup> in terms of size. To this end, we extend the Rényi divergence-based analysis in [FGdG<sup>+</sup>25] to the weak smoothness setting and present a security proof for FALCON<sup>ws</sup>. Building upon the improved analysis of the RO simulation, i.e. a tighter bound on the smoothing parameter of NTRU lattices (Theorem 3), together with a key lemma in [ZJR<sup>+</sup>25] (Lemma 6), we figure out the maximal  $\epsilon$  allowing for a formal proof solely on standard assumptions. As a result, we present new parameters for FALCON<sup>ws</sup> to support provable security on standard assumption while offering 17.8% (resp. 12.8%) smaller signature sizes compared to FALCON-512 (resp. FALCON-1024).

Theorem 5 characterizes the plain unforgeability of FALCON<sup>ws</sup>, in which the Rényi term  $r_p^{C_s} = 1$  due to the use of the rejection sampling.

**Theorem 5 (Unforgeability of FALCON<sup>ws</sup>).** *For any adversary  $\mathbf{A}$  against the UF-CMA security of FALCON<sup>ws</sup> running in time  $t_{\mathbf{A}}$ , making at most  $Q_s$  signing queries and  $Q_H$  RO queries, there exist adversary  $\mathbf{B}$  against  $(Q_H + 1)$ -R-ISIS running in time  $t_{\mathbf{B}} \approx t_{\mathbf{A}}$  such that for all  $C_s \in \mathbb{N}^{\geq 1}$  and  $a_u, a_p \in \mathbb{R}^{>1}$  it holds*

$$\begin{aligned} \text{Adv}_{\text{FALCON}^+, \mathbf{A}}^{Q_s\text{-UF-CMA}} &\leq \left( r_u^{C_s} \cdot \text{Adv}_{1,q,\alpha,\beta,\mathbf{B}}^{(Q_H+1)\text{-R-ISIS}} \right)^{\frac{a_u-1}{a_u}} \\ &\quad + \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp},\beta})^{C_s-i} (p_{\text{PreSmp},\beta})^i + \frac{Q_s(C_s + Q_H)}{2^k}, \end{aligned}$$

where

$$\begin{aligned} p_{\text{PreSmp},\beta} &= \max_{(\mathbf{B}, \cdot) \in \text{TpGen}, c \in \mathcal{R}_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(\mathbf{B}, s, (c, 0))} [\| (s_1, s_2) \|_2 \leq \beta], \\ r_u &= \max_{(\cdot, h) \in \text{TpGen}} R_{a_u}(\mathcal{P} \parallel \mathcal{Q}_h) \text{ with } \mathcal{P} = \mathcal{U}(\mathcal{R}_q) \text{ and } \mathcal{Q}_h \text{ the distribution of } \\ &\quad s_1 + s_2 \cdot h \bmod q, \text{ where } s_1, s_2 \sim D_{\mathcal{R},s} \\ r_p &= \max_{(\mathbf{B}, \cdot) \in \text{TpGen}, c \in \mathcal{R}_q} R_{a_p}(\text{PreSmp}(\mathbf{B}, s, (c, 0)) \parallel \mathcal{D}_{\Lambda(\mathbf{B}), s, (c, 0)}). \end{aligned}$$

As per Theorem 5, for the original FALCON<sup>ws</sup>-512 and FALCON<sup>ws</sup>-1024 proposed in [ZJR<sup>+</sup>25], the security loss becomes prohibitively large. To get provably secure parameters, it suffices to consider the Rényi loss  $r_u = R_{a_u}(\mathcal{U}(\mathcal{R}_q) \parallel \mathcal{Q}_h)$  from RO simulation. Lemma 6 shows a generic bound on the Rényi divergence.

**Lemma 6 ([ZJR<sup>+</sup>25]).** *Let  $\Lambda$  and  $\Lambda'$  be full-rank lattices with  $\Lambda' \subseteq \Lambda$ . For any  $a \in (1, \infty)$ ,  $\epsilon \in (0, 1)$ , and  $s \in \mathbb{R}$  such that  $\sqrt{2}s \geq \eta_\epsilon(\Lambda')$  and  $s \geq \sqrt{2} \cdot \eta_{\epsilon'}(\Lambda)$ , it holds that*

$$R_a(\mathcal{U}(\Lambda/\Lambda') \parallel D_{\Lambda, s, \mathbf{c}} \bmod \Lambda') \lesssim 1 + \frac{a(\epsilon + 4\epsilon')}{2}.$$

Applying Lemma 6 on  $\Lambda = \mathbb{Z}^{2n}$  and  $\Lambda_{h,q}$  along with  $\text{FALCON}^{\text{ws}}$  parameters, the term  $\epsilon'$  is sufficiently small to be safely ignored, and the corresponding values of  $\epsilon$  are computed to be  $2^{-82}$  for  $\text{FALCON}^{\text{ws}}\text{-512}$  and  $2^{-84}$  for  $\text{FALCON}^{\text{ws}}\text{-1024}$ , respectively. This results in a tighter bound for the Rényi divergence with respect to the RO simulation.

To optimize the overall Rényi loss term in Theorem 5, we propose to increase the value of  $\epsilon$  under the constraint that the resulting security loss remains bounded by a predefined threshold. Then, by leveraging the basis-specific smoothing parameter analysis introduced in Theorem 3, we are able to determine the actual (larger)  $\epsilon$  that equivalently satisfies the required condition from the previous step.

Table 7 shows the experimental results on 1,000 instances under different  $\epsilon$ . Since rejection sampling eliminates the security loss from the sampler and simultaneously enhances the concrete security of the underlying  $\mathcal{R}$ -ISIS problem, the forgery security maintains redundancy. Consequently, for  $\text{FALCON}^{\text{ws}}\text{-512}$  we select  $\epsilon = 2^{-23.5}$ , corresponding to a security loss of approximately 11 bits, while for  $\text{FALCON}^{\text{ws}}\text{-1024}$  we choose  $\epsilon = 2^{-21.5}$ , yielding a security loss of about 45 bits. These parameter sets allows for a formal security proof solely on standard assumptions.

**Table 7.** Bit loss  $\text{loss}_u$  from  $r_u^{C_s}$  for different  $\epsilon$ . Numbers are measured over 1,000 instances. The term  $(A, B)$  represents the maximum and mean of security loss respectively.

| $\epsilon$ of $\eta_\epsilon(\mathbb{Z}^{2n})$ | $\text{FALCON}^{\text{ws}}\text{-512}$ | $\text{FALCON}^{\text{ws}}\text{-1024}$ |
|------------------------------------------------|----------------------------------------|-----------------------------------------|
| $2^{-21.5}$                                    | (48.50, 24.20)                         | (44.60, 27.60)                          |
| $2^{-22.5}$                                    | (23.55, 11.56)                         | (21.68, 13.25)                          |
| $2^{-23.3}$                                    | (13.22, 6.41)                          | (10.54, 6.36)                           |

Table 8 shows our new parameter set for  $\text{FALCON}^{\text{ws}}$  with provable security. The provably secure  $\text{FALCON}^{\text{ws}}$  removes the additional non-standard assumption required by original  $\text{FALCON}^{\text{ws}}$ . Compared to the original  $\text{FALCON}$ , it offers a 20% (resp. 13.7%) smaller bandwidth (public key plus signature) for the NIST level I (resp. V).

**Table 8.** New parameter sets for FALCON<sup>ws</sup> with provable security.

| Parameter          | Description                          | NIST-I      | NIST-V      |
|--------------------|--------------------------------------|-------------|-------------|
| $n$                | Degree of ring $\mathcal{R}$         | 512         | 1024        |
| $q$                | Modulus                              | 2048        | 4096        |
| $\epsilon$         | Closeness parameter                  | $2^{-23.3}$ | $2^{-21.5}$ |
| $\sigma$           | Standard deviation                   | 58.11       | 81.21       |
| $\tau$             | Tailcut rate                         | 1.1         |             |
| $\beta$            | Max. signature norm bound            | 2045.45     | 4042.91     |
| $ \text{pk} $      | PK size in bytes                     | 704         | 1536        |
| $ \text{sig} $     | Estimated signature size in bytes    | 547         | 1115        |
| $\lambda$          | Key Rec. / Forg. security            | 134 / 128   | 273 / 295   |
| $a_p$              | Optimal Rényi order of sampler       | —           | —           |
| $a_u$              | Optimal Rényi order of RO simulation | 19.43       | 13.24       |
| $\text{loss}_p$    | Bits lost from $r_p^{C_s}$           | 0           |             |
| $\text{loss}_u$    | Bits lost from $r_u^{C_s}$           | 13          | 44.6        |
| Final bit security |                                      | 115         | 250         |

## 6 Refined Analysis of Strong Unforgeability

The analysis in [GJK24] indicates that FALCON offers no meaningful security guarantee for strong unforgeability. This limitation stems directly from that the original security proof relies on a reduction to the  $\mathcal{R}$ -SIS problem with a solution norm bound  $2\beta$ . This bound comes from the triangle inequality:

$$\|(s_1^* - s_1, s_2^* - s_2)\| \leq \|(s_1^*, s_2^*)\| + \|(s_1, s_2)\| \leq 2\beta$$

for the forgery signature  $(s_1^*, s_2^*)$  and the simulated signature  $(s_1, s_2)$ . While this bound is always correct, it is overly pessimistic and leads to severe consequences for concrete security:

- For FALCON<sup>+</sup>-512, this yields a security loss of approximately 25 bits compared to plain unforgeability (with bound  $\beta$ )
- For FALCON<sup>+</sup>-1024, the value  $2\beta$  exceeds the modulus  $q$ , rendering the corresponding  $\mathcal{R}$ -SIS instance trivial and the security guarantee meaningless.

This issue was resolved in the updated version [FGdG<sup>+</sup>25] by introducing a non-standard assumption. In contrast, we improve the concrete security analysis for strong unforgeability by deriving a tighter norm bound and a parameter optimization method.

### 6.1 A Tighter Norm Bound

We propose a refined, probabilistic analysis that replaces the worst-case bound of  $2\beta$  with a tighter, average-case bound of  $\sqrt{2}\beta$ . The rationale stems from the concentration of measure phenomena in high-dimensional spaces, which implies that randomly chosen vectors are, with high probability, nearly orthogonal.

**Lemma 7 (Lemma 1.5 of [Ban93]).** *For each  $c \geq (2\pi)^{-1/2}$ , one has*

1.  $\rho(L \setminus c\sqrt{n}B_n) < (c\sqrt{2\pi}e^{-\pi c^2})^n \rho(L)$
2.  $\rho((L + \mathbf{u}) \setminus c\sqrt{n}B_n) < 2(c\sqrt{2\pi}e^{-\pi c^2})^n \rho(L)$  for any  $\mathbf{u} \in \mathbb{R}^n$

where  $B_n$  denote the open unit ball in  $\mathbb{R}^n$ .

**Lemma 8 (Lemma 2.4 of [Ban95]).** *Let  $L$  be a lattice and let  $\mathbf{u}$  be an arbitrary vector in  $\mathbb{R}^n$ . Then*

$$\sum_{\substack{\mathbf{x} \in L + \mathbf{u} \\ \|\mathbf{x}\|_\infty \geq t}} e^{-\pi \|\mathbf{x}\|^2} < 2e^{-\pi t^2} \sum_{\mathbf{x} \in L} e^{-\pi \|\mathbf{x}\|^2}.$$

**Lemma 9.** *Let  $\mathcal{L} \subset \mathbb{R}^m$  be a lattice,  $s \geq \eta_\epsilon(\mathcal{L})$  with  $0 < \epsilon < 1$ ,  $\beta := \tau\sqrt{m}\frac{s}{\sqrt{2\pi}}$  and  $c > 0$  with  $\tau^2(c^2 - 1) > 1$ . Given  $\mathbf{w} \in \mathcal{L} + \mathbf{t}$  with  $\|\mathbf{w}\| \leq \beta$ , let  $\mathbf{v} \leftarrow D_{\mathcal{L}+\mathbf{t}, s}$ . Then for any  $0 < \alpha < \frac{\tau^2 \cdot (c^2 - 1) - 1}{2}$ ,*

$$\Pr[\|\mathbf{v} - \mathbf{w}\| \leq c\beta] \geq 1 - \left( 2e^{-\frac{\alpha^2 m}{2 \cdot \tau^2}} + 2 \left( ((c^2 - 1) \cdot \tau^2 - 2\alpha) e \right)^{m/2} \cdot e^{-\frac{((c^2 - 1) \cdot \tau^2 - 2\alpha)m}{2}} \right) \cdot \frac{1 + \epsilon}{1 - \epsilon}$$

*Proof.* For any  $t > 0$ , it holds that

$$\begin{aligned} \Pr[\|\mathbf{v} - \mathbf{w}\| \geq c \cdot \beta] &= \Pr[\|\mathbf{v} - \mathbf{w}\|^2 \geq c^2 \cdot \beta^2] \\ &= \Pr[\|\mathbf{v}\|^2 - 2\langle \mathbf{v}, \mathbf{w} \rangle + \|\mathbf{w}\|^2 \geq c^2 \cdot \beta^2] \\ &\leq \Pr[\|\mathbf{v}\|^2 - 2\langle \mathbf{v}, \mathbf{w} \rangle \geq (c^2 - 1) \cdot \beta^2] \\ &= \Pr[\|\mathbf{v}\|^2 - 2\langle \mathbf{v}, \mathbf{w} \rangle \geq (c^2 - 1) \cdot \beta^2 \wedge \langle \mathbf{v}, \mathbf{w} \rangle \leq -t] \\ &\quad + \Pr[\|\mathbf{v}\|^2 - 2\langle \mathbf{v}, \mathbf{w} \rangle \geq (c^2 - 1) \cdot \beta^2 \wedge \langle \mathbf{v}, \mathbf{w} \rangle \geq -t] \\ &\leq \Pr[\langle \mathbf{v}, \mathbf{w} \rangle \leq -t] + \Pr[\|\mathbf{v}\|^2 \geq (c^2 - 1) \cdot \beta^2 - 2t]. \end{aligned}$$

By Lemma 2.4 in [Ban95], it follows that

$$\begin{aligned} \Pr[\langle \mathbf{v}, \mathbf{w} \rangle \leq -t] &\leq \Pr[|\langle \mathbf{v}, \mathbf{w} \rangle| \geq t] \\ &\leq 2e^{-\pi \frac{t^2}{\|\mathbf{w}\|^2 s^2}} \cdot \frac{\rho_s(\Lambda)}{\rho_s(\Lambda + \mathbf{t})} \\ &\leq 2e^{-\pi \frac{t^2}{\|\mathbf{w}\|^2 s^2}} \cdot \frac{1 + \epsilon}{1 - \epsilon}. \end{aligned}$$

By Lemma 1.5 in [Ban93], for  $(c^2 - 1) \cdot \beta^2 - 2t > \frac{ms^2}{2\pi}$  it follows that

$$\begin{aligned} \Pr[\|\mathbf{v}\|^2 \geq (c^2 - 1) \cdot \beta^2 - 2t] &\leq 2C^m \cdot \frac{\rho_s(\Lambda)}{\rho_s(\Lambda + \mathbf{w})} \\ &\leq 2C^m \cdot \frac{1 + \epsilon}{1 - \epsilon}. \end{aligned}$$

with  $C = \sqrt{\frac{(c^2 - 1) \cdot \beta^2 - 2t}{ms^2}} \sqrt{2\pi e} \cdot e^{-\pi \frac{(c^2 - 1) \cdot \beta^2 - 2t}{ms^2}}$ . By setting  $t = \alpha \cdot \frac{ms^2}{2\pi}$  for any  $0 < \alpha < \frac{\tau^2 \cdot (c^2 - 1) - 1}{2}$ , then

$$\begin{aligned} \Pr[\|\mathbf{v} - \mathbf{w}\| \geq c \cdot \beta] &\leq \Pr[\langle \mathbf{v}, \mathbf{w} \rangle \leq -t] + \Pr[\|\mathbf{v}\|^2 \geq (c^2 - 1) \cdot \beta^2 - 2t] \\ &\leq \left( 2e^{-\pi \frac{t^2}{\|\mathbf{w}\|^2 s^2}} + 2C^m \right) \cdot \frac{1 + \epsilon}{1 - \epsilon} \\ &\leq \left( 2e^{-\frac{\alpha^2 m}{2 \cdot \tau^2}} + 2 \left( ((c^2 - 1) \cdot \tau^2 - 2\alpha) e \right)^{m/2} \cdot e^{-\frac{((c^2 - 1) \cdot \tau^2 - 2\alpha)m}{2}} \right) \cdot \frac{1 + \epsilon}{1 - \epsilon}. \end{aligned}$$

□

**Theorem 6 (Strong Unforgeability, adapted from [GJK24]).** *For any adversary  $\mathbf{A}$  against the **SUF-CMA** security of  $\text{FALCON}^+$  running in time  $t_{\mathbf{A}}$ , making at most  $Q_s$  signing queries and  $Q_H$  RO queries, there exist adversary  $\mathbf{B}$  against  $\mathcal{R}$ -**SIS** running in time  $t_{\mathbf{B}} \approx t_{\mathbf{A}}$  such that for all  $C_s \in \mathbb{N}^{\geq 1}$ ,  $L \in [Q_H + 1]$  and  $a_u, a_p \in \mathbb{R}^{>1}$  it holds*

$$\begin{aligned} \text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}} &\leq \frac{(Q_H + 1)}{L \cdot p_{\text{PreSmp}, \beta}} \left( r_u^{C_s + L} \cdot \left( r_p^{C_s + 1} \cdot \left( \frac{1}{t} \cdot \text{Adv}_{1, q, \alpha, c, \beta, \mathbf{B}}^{\mathcal{R}\text{-SIS}} + \frac{1}{2^{n-1}} \right) \right)^{\frac{a_p - 1}{a_p}} \right)^{\frac{a_u - 1}{a_u}} \\ &\quad + \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp}, \beta})^{C_s - i} (p_{\text{PreSmp}, \beta})^i + \frac{Q_s(C_s + Q_H)}{2^k}, \end{aligned}$$

where

$t$  is the lower bound established in Lemma 9,

$$\begin{aligned} p_{\text{PreSmp}, \beta} &= \max_{(\mathbf{B}, \cdot) \in \text{TpdGen}, c \in \mathcal{R}_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(\mathbf{B}, s, (c, 0))} [\|(s_1, s_2)\|_2 \leq \beta], \\ r_u &= R_{a_u}(\mathcal{P} \parallel \mathcal{Q}_h) \text{ with } \mathcal{P} = \mathcal{U}(\mathcal{R}_q) \text{ and } \mathcal{Q}_h \text{ the distribution of } s_1 + s_2 \cdot h \\ &\text{mod } q, \text{ where } s_1, s_2 \sim D_{\mathcal{R}, s} \text{ and } (\cdot, h) \in \text{TpdGen} \\ r_p &= R_{a_p}(\text{PreSmp}(\mathbf{B}, s, (c, 0)) \parallel \mathcal{D}_{\Lambda(\mathbf{B}), s, (c, 0)}) \text{ with } (\mathbf{B}, \cdot) \in \text{TpdGen}. \end{aligned}$$

*Proof sketch.* The proof largely follows Theorem 1 in [GJK24], with two modifications. First, the Rényi divergences  $r_p$  and  $r_u$  are defined relative to the specific public key  $h$ , rather than using their upper bounds as in [GJK24]. Second, we adjust the reduction from the  $\mathcal{R}$ -SIS problem to Game  $G_8$  in the final step. The complete proof is provided in Appendix A.1.

*Remark 5.* If we fix  $\tau = 1.1$  and  $c = \sqrt{2}$ , then for FALCON-512,  $\epsilon = 2^{-35.5}$ ,  $\alpha = 0.055$  and the computed probability lower bound is 0.2627, meaning at most 2 bits of security loss for our tighter norm bound  $\sqrt{2}\beta$ . Similarly, for FALCON-1024,  $\epsilon = 2^{-36}$ ,  $\alpha = 0.062$  and the computed probability lower bound is 0.867, meaning at most 1 bits of security loss. With this new bound, we can obviously improve the SUF-CMA security. More importantly, the new bound satisfies  $\sqrt{2}\beta < q$ , which renders the strong unforgeability guarantee for FALCON-1024 meaningful. The estimated SUF-CMA security of FALCON with bound  $\sqrt{2}\beta$  is shown in Table 9.

**Table 9.** SUF-CMA security of FALCON.

|         | FALCON-512 | FALCON-1024 |
|---------|------------|-------------|
| SUF-CMA | 107        | 251         |

*Remark 6.* Building upon the refined Rényi divergence analysis established in Section 4, we demonstrate that increasing  $\epsilon$  can also enhance the SUF-CMA security of FALCON. However, even with our improved analysis in Theorem 2, the Rényi loss term  $r_p$  grows rapidly with  $\epsilon$ , limiting the extent to which  $\epsilon$  can be increased. Consequently, the resulting security improvement remains modest, where  $\epsilon \approx 2^{-32}$ . In the next subsection, we present a FALCON<sup>ws</sup> variant specifically optimized for SUF-CMA security. Since its sampler introduces no Rényi loss, this variant offers greater flexibility in parameter adjustment,

## 6.2 New FALCON<sup>ws</sup> Parameters with SUF-CMA Security

Directly tightening the norm bound from  $2\beta$  to  $\sqrt{2}\beta$  already provides an obvious enhancement for FALCON-type signature’s strong unforgeability. In this subsection we present a new parameter set for FALCON<sup>ws</sup> with SUF-CMA security.

In addition to the refined Rényi divergence analysis established in Section 4 and the tighter norm bound of  $\sqrt{2}\beta$ , we further observe that the Rényi loss in Theorem 6 is not optimal for FALCON parameters. Specifically, we can further tighten the reduction by co-optimizing  $\epsilon$  and the auxiliary parameter  $L$ , a technique originating from [BRTZ24] and employed in [FGdG<sup>+</sup>25], for which we present an optimized strategy for its selection.

Our approach aims to balance the multiplicative factor in the security reduction:

$$\frac{Q_H}{L} \cdot r_u^{C_s+L} \cdot r_p^{C_s}.$$

By carefully increasing  $\epsilon$ , we drive this factor close to  $Q_H$ , thereby ensuring that the first and last terms in the advantage expression  $\text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}}$  become nearly optimal under concrete parameter settings. This optimization allows us to achieve a tight security proof while maintaining practical signature size and security levels, as demonstrated in our experimental evaluation in Table 10.



**Table 10.** A FALCON<sup>ws</sup> variant targeting SUF-CMA security.

| Parameter          | Description                          | NIST-I      | NIST-V    |
|--------------------|--------------------------------------|-------------|-----------|
| $n$                | Degree of ring $\mathcal{R}$         | 512         | 1024      |
| $q$                | Modulus                              | 4096        | 8192      |
| $\epsilon$         | Closeness parameter                  | $2^{-21.5}$ |           |
| $\sigma$           | Standard deviation                   | 79.993      | 114.86    |
| $\tau$             | Tailcut rate                         | 1.1         |           |
| $\sqrt{2}\beta$    | Max. signature norm bound            | 3982.1      | 8085.8    |
| $ \mathbf{pk} $    | PK size in bytes                     | 768         | 1664      |
| $ \mathbf{sig} $   | Estimated signature size in bytes    | 576         | 1179      |
| $\lambda$          | Key Rec. / Strong Forg.              | 134 / 114   | 273 / 265 |
| $L$                | Number of programmed RO queries      | $2^{55}$    | $2^{56}$  |
| $a_p$              | Optimal Rényi order of sampler       | —           | —         |
| $a_u$              | Optimal Rényi order of RO simulation | 5.23        | 12.55     |
| $loss_p$           | Bits lost from $r_p^{C_s}$           | 0           |           |
| $loss_u$           | Bits lost from $r_u^{C_s}$           | 45.52       | 42.29     |
| Final bit security |                                      | 114         | 265       |

*Remark 7.* It should be noted that although the security loss  $loss_u$  for both parameter sets significantly exceeds 1 bit, we emphasize that this reduction loss does not ultimately compromise the overall security guarantee. This is because the multiplicative prefactor  $Q_H/L$  of the first term in  $\text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}}$  effectively compensates for approximately  $\log(L) = 55$  or 56 bits of security. That is, through careful parameter tuning, we ensure that the value of the first term in the advantage expression  $\text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}}$  remains strictly smaller than the third one, thereby rendering the observed reduction loss inconsequential to the final security level.

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## Appendix A

### A.1 Proof of Theorem 6

**Lemma 10 (Min-Entropy of Gaussian [GJK24]).** *Let  $n \in \mathbb{N}$ ,  $\Lambda = \mathbb{Z}^n$ ,  $\epsilon \in (0, 1/2)$  and  $s \geq \eta_\epsilon(\Lambda) > 2$ . Then for any  $c \in \mathbb{R}^n$  and  $x \in \Lambda$ ,  $D_{\Lambda, s, c}(\mathbf{x}) \leq \frac{1}{2^{n-1}}$ .*

**Theorem 6 (Strong Unforgeability, adapted from [GJK24]).** *For any adversary  $\mathbf{A}$  against the **SUF-CMA** security of  $\text{FALCON}^+$  running in time  $t_{\mathbf{A}}$ , making at most  $Q_s$  signing queries and  $Q_H$  RO queries, there exist adversary  $\mathbf{B}$  against **R-SIS** running in time  $t_{\mathbf{B}} \approx t_{\mathbf{A}}$  such that for all  $C_s \in \mathbb{N}^{\geq 1}$ ,  $L \in [Q_H + 1]$  and  $a_u, a_p \in \mathbb{R}^{>1}$  it holds*

$$\begin{aligned} \text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}} &\leq \frac{(Q_H + 1)}{L \cdot p_{\text{PreSmp}, \beta}} \left( r_u^{C_s + L} \cdot \left( r_p^{C_s + 1} \cdot \left( \frac{1}{t} \cdot \text{Adv}_{1, q, \alpha, c\beta, \mathbf{B}}^{\mathcal{R}\text{-SIS}} + \frac{1}{2^{n-1}} \right) \right)^{\frac{a_p - 1}{a_p}} \right)^{\frac{a_u - 1}{a_u}} \\ &\quad + \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp}, \beta})^{C_s - i} (p_{\text{PreSmp}, \beta})^i + \frac{Q_s(C_s + Q_H)}{2^k}, \end{aligned}$$

where

$t$  is the lower bound established in Lemma 9,

$p_{\text{PreSmp}, \beta} = \max_{(\mathbf{B}, \cdot) \in \text{TpdGen}, c \in \mathcal{R}_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(\mathbf{B}, s, (c, 0))} [\|(s_1, s_2)\|_2 \leq \beta]$ ,  
 $r_u = R_{a_u}(\mathcal{P} \parallel \mathcal{Q}_h)$  with  $\mathcal{P} = \mathcal{U}(\mathcal{R}_q)$  and  $\mathcal{Q}_h$  the distribution of  $s_1 + s_2 \cdot h \bmod q$ , where  $s_1, s_2 \sim D_{\mathcal{R}, s}$  and  $(\cdot, h) \in \text{TpdGen}$   
 $r_p = R_{a_p}(\text{PreSmp}(\mathbf{B}, s, (c, 0)) \parallel \mathcal{D}_{\Lambda(\mathbf{B}), s, (c, 0)})$  with  $(\mathbf{B}, \cdot) \in \text{TpdGen}$ .

*Proof.* Consider the sequence of the following games.

*Game  $G_0$ .* This is the strong unforgeability game for  $\text{COREFALCON}^+$  so by definition we have

$$\Pr[G_0^{\mathbf{A}} \Rightarrow 1] = \text{Adv}_{\text{COREFALCON}^+, \mathbf{A}}^{Q_s\text{-SUF-CMA}}.$$

*Game  $G_1$ .* This game is identical to the previous one, except that it aborts if the overall number of sampled preimages in the signing oracle, i.e. including potential repetitions, exceeds threshold  $C_s$ .

*Claim.* For  $p_{\text{PreSmp}, \beta} = \max_{c \in \mathcal{R}_q} \Pr_{(s_1, s_2) \leftarrow \text{PreSmp}(\mathbf{B}, s, (c, 0))} [\|(s_1, s_2)\|_2 \leq \beta]$  it holds that

$$|\Pr[G_0^{\mathbf{A}} \Rightarrow 1] - \Pr[G_1^{\mathbf{A}} \Rightarrow 1]| \leq \sum_{i=0}^{Q_s} \binom{C_s}{i} (1 - p_{\text{PreSmp}, \beta})^{C_s - i} (p_{\text{PreSmp}, \beta})^i.$$

*Proof.* To proof the claim, we model the experiment using a binomial distributed random variable  $X \sim B(C_s, p_{\text{PreSmp}, \beta})$ , i.e. we have  $C_s$  Bernoulli trials and success probability  $p_{\text{PreSmp}, \beta}$ . A trial corresponds to sampling a preimage using  $\text{PreSmp}$  in the signing oracle and the trial succeeds if the norm is sufficiently small, i.e.  $\|(s_1, s_2)\|_2 \leq \beta$ . Hence, the random variable, counting the overall number of successes in the Bernoulli trials, tells us the number of signing queries we are able to answer. Since we need to answer  $Q_s$  signing queries, we are interested in the CDF for value  $Q_s$ , i.e.  $\Pr[X < Q_s]$  which is exactly the claim.  $\square$

*Game  $G_2$ .* This game is identical to the previous one except that it aborts in the signing oracle  $\text{Sgn}$  if there already exists a query to the RO on the same public key, salt  $r$ , and message  $m$ .

*Claim.*  $|\Pr[G_1^{\mathbf{A}} \Rightarrow 1] - \Pr[G_2^{\mathbf{A}} \Rightarrow 1]| \leq \frac{Q_s(C_s + Q_H)}{2^k}$ .

*Proof.* The salt  $r$  is chosen uniformly at random from  $\{0,1\}^k$  for each signing query. The total number of elements in  $\mathcal{H}$  is at most  $C_s + Q_H$ , as at most one element is added per query to  $H$  and there are  $C_s$  implicit queries via  $\text{Sgn}$  and  $Q_H$  direct ones. Thus, the probability that the freshly chosen salt was part of a previous query is at most  $\frac{C_s + Q_H}{2^k}$ . For  $Q_s$  queries to the signing oracle  $\text{Sgn}$ , we obtain the claimed bound.  $\square$

*Game  $G_3$ .* This game is identical to the previous one except that the game computes a preimage of  $c^*$  with respect to  $h$  using the preimage sampler  $\text{PreSmp}$  with trapdoor  $\mathbf{B}$ . If the norm of the resulting preimage  $(s'_1, s'_2)$  is larger than  $\beta$ , the game aborts.

*Claim.*  $\Pr[G_2^A \Rightarrow 1] = (p_{\text{PreSmp}, \beta})^{-1} \cdot \Pr[G_3^A \Rightarrow 1]$

*Proof.* The probability that the abort does not trigger is  $p_{\text{PreSmp}, \beta}$ . Since the preimage is computed after adversary  $A$  outputs their forgery and the winning condition is not affected by  $(s'_1, s'_2)$ , the abort event is independent of  $A$ 's winning probability. Hence we can apply the multiplicative difference lemma to obtain the statement.  $\square$

*Game  $G_4$ .* This game is identical to the previous one except for the following changes. For a fixed integer  $L \in [Q_H + 1]$ , we choose a random subset  $S \subset [Q_H]$  with  $|S| = L$ . For every query to  $H$ , we record the query number  $l$  and the game abort if the the RO query of the forgery corresponds to a query to  $H$  and the query number  $l^*$  is not in  $S$ .

*Claim.*  $\Pr[G_3^A \Rightarrow 1] = \frac{Q_H + 1}{L} \cdot \Pr[G_4^A \Rightarrow 1]$

*Proof.* First note that the probability of an abort is independent of the winning probability in  $G_3$  since set  $S$  is not used anywhere else and hence does not influence the winning probability. The game does not abort if the challenge RO query corresponds to a signing query but only to a direct RO query. Since  $S$  is of fixed size  $L$ , we can apply the multiplicative difference lemma and obtain the claim.  $\square$

*Game  $G_5$ .* This game is the same as the previous one, except that the RO no longer returns a uniformly random element  $c \leftarrow \mathcal{R}_q$ . Instead, it computes  $c$  as follows: It samples two elements  $s_1, s_2$  from  $D_{\mathcal{R}, s}$ . Then,  $c$  is computed as  $c = s_1 + s_2 \cdot h \pmod q$ , where  $h$  is the public key.

*Claim.* For  $h \in \mathcal{R}$ , let  $\mathcal{P} = \mathcal{U}(\mathcal{R}_q)$  and  $\mathcal{Q}_h$  be the distribution of  $s_1 + s_2 \cdot h \pmod q$  where  $s_1, s_2 \leftarrow D_{\mathcal{R}, s}$ . Then, for any  $a_u \in (1, \infty)$ ,

$$\Pr[G_4^A \Rightarrow 1] \leq (R_{a_u}(\mathcal{P} \parallel \mathcal{Q}_h)^{C_s} \cdot \Pr[G_5^A \Rightarrow 1])^{\frac{a_u - 1}{a_u}}$$

*Proof.* Define two underlying distributions for a  $(Q+1)$ -tuple of random variables  $(c_0 = (\mathbf{B}, h), c_1, \dots, c_Q)$ .

$$\begin{array}{ll}
\bar{\mathcal{P}} & \bar{\mathcal{Q}} \\
(\mathbf{B}, h) \xleftarrow{s} \text{TpdGen} & (\mathbf{B}, h) \xleftarrow{s} \text{TpdGen} \\
\text{for } i \in [Q] & \text{for } i \in [Q] \\
\quad c_i \xleftarrow{s} \mathcal{R}_q & \quad (s_1, s_2) \leftarrow D_{\mathcal{R}, s} \\
\text{return } ((\mathbf{B}, h), c_1, \dots, c_Q) & \quad c_i = s_1 + s_2 \cdot h \pmod{q} \\
& \text{return } ((\mathbf{B}, h), c_1, \dots, c_Q)
\end{array}$$

These distributions describe the underlying distributions of  $G_4$  and  $G_5$ . By the data processing inequality it holds that, for any  $a \in (1, \infty)$ ,

$$R_a(G_4 \parallel G_5) \leq R_a(\bar{\mathcal{P}} \parallel \bar{\mathcal{Q}}).$$

Let the marginal distribution of  $c_i$  be denoted by  $\bar{\mathcal{P}}_i$  ( $\bar{\mathcal{Q}}_i$  resp.) and the distribution of  $c_i$  conditioned on  $c_{<i} = (c_0, \dots, c_{i-1})$  as  $\bar{\mathcal{P}}_{i|c_{<i}}$  ( $\bar{\mathcal{Q}}_{i|c_{<i}}$  resp.). Since the distribution of  $c_0 = (\mathbf{B}, h)$  is the same for  $\bar{\mathcal{P}}$  and  $\bar{\mathcal{Q}}$ , it holds that

$$R_a(\bar{\mathcal{P}}_0 \parallel \bar{\mathcal{Q}}_0) = 1.$$

For the conditional distributions, note that random variable  $c_i$  is independent of the previous random variables  $c_1, \dots, c_{i-1}$ . However,  $c_i$  might depend on  $h$  and thus on random variable  $c_0 = (\mathbf{B}, h)$ . Hence for all  $i \in [Q+1]$ ,

$$\begin{aligned}
R_a(\bar{\mathcal{P}}_{i|c_{<i}} \parallel \bar{\mathcal{Q}}_{i|c_{<i}}) &= R_a(\bar{\mathcal{P}}_{i|(c_0, \dots, c_{i-1})} \parallel \bar{\mathcal{Q}}_{i|(c_0, \dots, c_{i-1})}) \\
&= R_a(\mathcal{P} \parallel \bar{\mathcal{Q}}_{i|c_0}) \\
&= R_a(\mathcal{P} \parallel \mathcal{Q}_h),
\end{aligned}$$

where  $\mathcal{P} = \mathcal{U}(\mathcal{R}_q)$  and  $\mathcal{Q}_h$  the distribution of  $s_1 + s_2 \cdot h \pmod{q}$  where  $s_1, s_2 \leftarrow D_{\mathcal{R}, s}$ . Note that  $h$  does not occur in distribution  $\mathcal{P}$  because the individual random variables  $c_i$  (for  $i \geq 1$ ) are independent of  $h$ . It then follows that

$$R_a(\bar{\mathcal{P}} \parallel \bar{\mathcal{Q}}) \leq R_a(\mathcal{P} \parallel \mathcal{Q}_h)^Q.$$

Finally, we obtain

$$\Pr[G_5^A \Rightarrow 1] \geq \frac{\Pr[G_4^A \Rightarrow 1]^{\frac{a}{a-1}}}{R_a(G_4 \parallel G_5)} \geq \frac{\Pr[G_4^A \Rightarrow 1]^{\frac{a}{a-1}}}{R_a(\mathcal{P} \parallel \mathcal{Q}_h)^Q}.$$

The claim follows by setting  $Q = C_s$  due to at most  $C_s$  queries from **Sgn** to the RO.

*Game  $G_6$ .* This game is identical to the previous one except that the output of the preimage sampler  $\text{PreSmp}(\mathbf{B}, s, (c, 0))$  is replaced by a Gaussian over the same lattice, standard deviation and center, namely  $D_{\Lambda(\mathbf{B}), s, (c, 0)}$ .

*Claim.* For distributions  $\text{PreSmp} = \text{PreSmp}(\mathbf{B}, s, (c, 0))$ ,  $D = D_{\Lambda(\mathbf{B}), s, (c, 0)}$ , and  $a_p \in (1, \infty)$  it holds that

$$\Pr[G_5^A \Rightarrow 1] \leq (R_{a_p}(\text{PreSmp} \parallel D)^{C_s} \cdot \Pr[G_6^A \Rightarrow 1])^{\frac{a_p-1}{a_p}}.$$

*Proof.* The claim follows analogously to Game  $G_5$ .

*Game  $G_7$ .* This game is identical to the previous one except that  $s_1, s_2$  are not sampled from a Gaussian distribution centred at  $(c, 0)$  as before. Instead, the preimage of  $c$  that was sampled in the RO is used.

*Claim.*  $\Pr[G_6^A \Rightarrow 1] = \Pr[G_7^A \Rightarrow 1]$ .

*Proof.* We need to show that the distributions of the games are the same. The RO output  $c$  is the same in both games. In  $G_6$ , the signing oracle outputs  $(s_1, s_2) \sim D_{\Lambda(\mathbf{B}), s, (c, 0)}$ . Since  $\Lambda(\mathbf{B})$  is the NTRU lattice for  $h$  and  $q$  and the distribution is shifted by  $(c, 0)$  the output is distributed according to a Gaussian  $D_{\mathcal{R}, s}$  conditioned on  $s_1 + s_2 \cdot h = c \pmod q$ . The output distribution in Game  $G_7$  is a Gaussian  $D_{\mathcal{R}, s}$  as well where the condition  $s_1 + s_2 \cdot h = c \pmod q$  is fulfilled by construction.

*Game  $G_8$ .* This game is identical to the previous one except that the game aborts if  $s_1^*, s_2^*$  corresponding to the adversary's output equals the preimage with which the challenge RO output  $c^*$  was computed.

*Claim.*  $|\Pr[G_7^A \Rightarrow 1] - \Pr[G_8^A \Rightarrow 1]| \leq \frac{1}{2^{n-1}}$ .

*Proof.* We distinguish two cases: first, the adversary queried the signing oracle corresponding to  $c^*$ , i.e., a signing query to  $m^*$  output  $(r^*, \cdot)$  and  $H(h, r^*, m^*) = c^*$ . In this case, the abort event cannot trigger because if the preimages are the same, the signature must be the same as well. Second, if the signing oracle was not queried corresponding to  $c^*$ , the adversary does not have any information about the preimages of  $c^*$  except that they are Gaussian distributed. Hence, we obtain the claimed upper bound by using the min-entropy of a sample from a Gaussian distribution conditioned on  $c^*$  from Lemma 10.

*Reduction from  $\mathcal{R}$ -SIS to  $G_8$*  We now can reduce  $\mathcal{R}$ -SIS to Game  $G_8$ .

*Claim.* There exists an adversary  $B$  against  $\mathcal{R}$ -SIS such that

$$\Pr[G_8^A \Rightarrow 1] \leq \frac{1}{t} \cdot \text{Adv}_{1, q, \alpha, c\beta, B}^{\mathcal{R}\text{-SIS}},$$

where  $t$  is the lower bound established in Lemma 9.

*Proof.* Due to the changes in the previous games, adversary  $B$  can perfectly simulate the game for adversary  $A$  against  $G_8$  without having the secret key for  $h$ . Assume that  $A$  wins the strong unforgeability game  $G_8$ , i.e. the forgery verifies and the tuple  $(m^*, \text{sig}^*)$  was not queried before. It implies that the output  $(s_1^* - s'_1, s_2^* - s'_2)$  of adversary  $B$  fulfills the following conditions. First, it holds that

$$(s_1^* - s'_1) + (s_2^* - s'_2)h = 0.$$

Second, it cannot equal 0 due to the changes in  $G_8$ . Third, by Lemma 9,  $\Pr[E] > t$ , where  $E$  denote the event that  $\|(s_1^* - s'_1, s_2^* - s'_2)\| \leq c\beta$ . Then we have

$$\begin{aligned}
\text{Adv}_{1,q,\alpha,c\beta,\mathbf{B}}^{\mathcal{R}\text{-}\mathbf{SIS}} &= \Pr[\mathbf{B} \text{ solves } \mathcal{R}\text{-}SIS] \\
&= \Pr[G_8^{\mathbf{A}} \Rightarrow 1 \wedge E] \\
&= \Pr[G_8^{\mathbf{A}} \Rightarrow 1] \cdot \Pr[E] \\
&\geq \Pr[G_8^{\mathbf{A}} \Rightarrow 1] \cdot t.
\end{aligned}$$

This completes the proof. □